



Technical Campus

योग: कर्मसु कौशलम्
IN PURSUIT OF PERFECTION



**SCHOOL OF ENGINEERING
AND TECHNOLOGY**

Process Automation (IOT310)

By: Dr. Divya Agarwal

Course Overview:

- Process and Industrial Automation specialization offers comprehensive knowledge and professional level skills focused on developing and deploying software robots. It starts with the basic concepts of Robotic Process Automation. In the present global scenario, industries are moving towards complete automation. Hence, this course is foundation of engineers who want to specialize in industrial and process automation field.

Course Objective:

- To understand the Fundamental and basic principles of Robotics Process Automation, Applications in various Industries.
- To install the UiPath Studio and understand difference between different versions
- To use different activities while using excel sheet and automate email related activities.
- To understand benefits of given industrial automation systems, describe their functions and compare characteristics of given automation systems.

Course Outcomes (CO):

- **CO1** Able to Map and assess some of the business processes that are fit for automation.
- **CO2** Able to connect to UiPath Automation Cloud, download the installer, and set up their own attended automation environment and Build an automation using StudioX.
- **CO3** Able to effectively automate tasks involving use of Excel files, adapt the automation project to dynamic Excel ranges.
- **CO4** Able to apply and map realtime application of industrial automation

- **UNIT I**

❑ **Introduction to process automation:** Definition of automation, Socio economic impacts of automation, Types of Automation, Introduction to Robot Process Automation, How is Automation Driving the Digital Transformation? Role of Automation in Business; Build first automation process with UiPath StudioX

Syllabus Contents

- **UNIT II**

- **Automation using StudioX:** computational concepts when building an automation project, Identify the tasks that are suitable for automation, breakdown a task and document it in a RobotPath. Introduction to file and folder automation,
- **Introduction to UI Automation:** Introduction to UI automation, recording UI interactions, using the Object Repository, UI automation activities, extracting data from an application, using of different control flow activities and its use

Syllabus Contents

- **UNIT III**
- **Error Handling:** Use StudioX tools to validate and analyze automation projects for handling and troubleshooting errors, build automations using best practices that increase reusability and readability.
- **Email Automation:** Use actions and resources related to email automation, create automation projects using StudioX email specific activities, create email content using both text and HTML options.
- **Excel Automation:** Automate tasks involving use of Excel files such as usage of cell activities, range activities, pivot activities, chart activities and workbook activities.

Syllabus Contents



- **UNIT IV**
- **Introduction to Industrial Automation:** Need and benefits of Industrial Automation, Automation Hierarchy, Basic components of Automation systems, description of each component, types of automation systems flexible, fixed and programmable systems
- **The Future Automated Factory:** Future Automated Factory, Human Workers in the Future Automated Factory, The social impact

- **UNIT I**

❑ **Introduction to process automation:** Definition of automation, Socio economic impacts of automation, Types of Automation, Introduction to Robot Process Automation, How is Automation Driving the Digital Transformation? Role of Automation in Business; Build first automation process with UiPath StudioX

Introduction to Automation

- Automation involves using technology to replicate human actions, often repetitive or rulebased tasks.
- Automation refers to the use of technology, machinery, or systems to perform tasks or processes with minimal human intervention.
- The primary goal of automation is to increase efficiency, reduce manual labor, and enhance accuracy in the execution of repetitive or complex operations.
- Automation can be applied across various industries and domains, ranging from manufacturing and business processes to information technology and household chores.

Key Characteristics of Automation

- **Efficiency:** Automation aims to streamline processes, often leading to increased productivity and faster task execution compared to manual methods.
- **Accuracy:** Automated systems are designed to perform tasks with a high degree of precision, minimizing the risk of errors associated with human involvement.
- **Consistency:** Automation ensures consistent performance, adhering to predefined rules and instructions without deviations over time.
- **Speed:** Automated processes can operate at a much faster pace than human counterparts, enabling quick and timely completion of tasks.

Key Characteristics of Automation

- **Cost Savings:** While there may be an initial investment in setting up automation systems, the long-term benefits often include reduced labor costs and increased overall operational cost-effectiveness.
- **Scalability:** Automated systems can easily scale to handle increased workloads or variations in demand without a proportional increase in human resources.
- **Reliability:** Well-designed automation systems are reliable and can operate continuously, minimizing downtime and disruptions.
- **Safety:** Automation can be applied in environments that are hazardous or challenging for humans, improving overall safety by reducing exposure to potential risks.

Types of Automation

- **Industrial Automation:** Applied in manufacturing and production processes, involving the use of control systems, machinery, and robots to handle tasks like assembly, packaging, and quality control.
- **Business Process Automation (BPA):** Involves automating repetitive, rule based tasks in business workflows, such as data entry, document processing, and customer support.
- **IT Process Automation (ITPA):** Automates routine tasks in IT operations, including system monitoring, software deployment, and troubleshooting.

Types of Automation

- **Home Automation:** Applies to smart homes, where devices and systems are interconnected and can be controlled remotely for tasks like climate control, security, and entertainment.
- **Robotic Process Automation (RPA):** Focuses on automating routine, rule based tasks using software robots to interact with applications and systems in a manner similar to human users.
- **Test Automation:** Involves using automated scripts to perform software testing tasks, ensuring the functionality and reliability of applications.

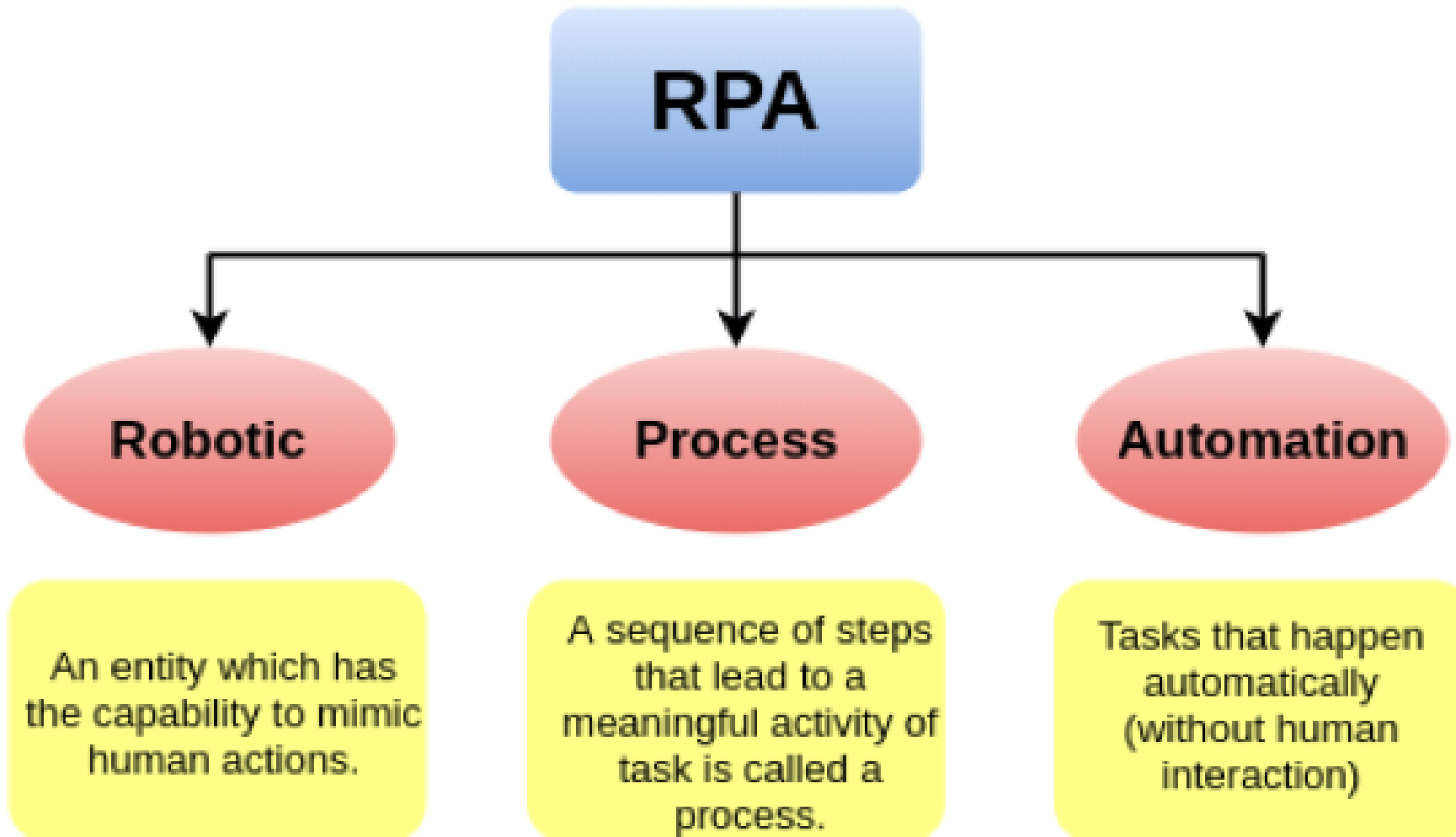
What is RPA?

- RPA stands for **Robotic Process Automation**.
- It is the technology used for software tools that automate human tasks, which are manual, rule based, or repetitive.
- Typically, it is like a bot that performs such tasks at a much higher rate than a human alone.
- These RPA software bots never sleep and make zero mistakes, and can interact with inhouse applications, websites, user portals, etc.
- They can log into applications, enter data, open emails and attachments, calculate and complete tasks, and then log out.
- 'Robot' in 'RPA' is not a physical robot but a virtual system that helps in automating the repetitive manual computing or business process tasks.

What RPA does?

- Mimics human interaction with computer systems (clicking, typing, navigating, extracting data).
- Automates repetitive tasks across different applications and platforms.
- Follows predefined rules and instructions to complete tasks consistently and accurately.

What is RPA?



Why RPA?

- Robotic Process Automation is economically capable as compared to any other automation solutions.
- It is the new buzz word in the IT industry. It has shifted the traditional way of doing the business task manually into an automatic task within an organization.
- RPA technology uses bots that interact with web applications, web sites, excel worksheets, and emails to automate the tasks just like a human.

BENEFITS OF RPA

- **Cost Savings:** RPA helps organizations to save a huge amount of cost as it is typically cheaper than hiring an employee to perform the same set of tasks.
- **Less Error:** RPA works on standard logic and does not get bored, distracted, or tired. Hence, the probability of making errors reduces to a great extent, which means less rework and an enhanced reputation for efficiency.
- **Faster Processing:** RPA works faster than human employees as computer software does not need breaks, food, rest, etc., and can perform repetitive operations tirelessly. With RPA, processing time becomes predictable and consistent, which ensures highquality customer service across the operations.
- **Better Regulatory Compliance:** RPA software works on the logic and data fed to it and does what is only needed as per the given instructions. Hence, there are minimal chances of not complying with the standard regulations.

- **Better Customer Service:** When RPA is implemented in a business, it frees many of its employees who can spend their time working on customer related services. It is very beneficial for businesses that receive a lot of customer queries. It also leads to increased productivity for employees.
- **Auditable & Secure:** RPA bots will only access the data for which they are given permission and create a detailed audit trail of all activity.
- **Low Technical Barrier:** RPA does not require any programming skills to configure the software robot. Since it is a codefree technology, any nontechnical person can set up the bot using draganddrop features. It also includes the 'Recorder' to record the steps of automation.

1) Attended RPA (which may be referred to as robotic desktop automation or RDA):

- First form of RPA that emerged, back in 2003 or so.
- Attended RPA means that the software provides collaboration with a person for certain tasks.
- Example: would be in the call center, where a rep can have the RPA system handle looking up information while he or she talks to a customer.

2) Unattended RPA: Second generation of RPA.

- With unattended RPA, you can automate a process without the need for human involvement – that is, the bot is triggered when certain events happen,
- Example: such as when a customer emails an invoice.
- Consider that unattended RPA is generally for backoffice functions.

3) Intelligent process automation or IPA (this may also be referred to as cognitive RPA):

- This is the latest generation of RPA technology, which leverages AI to allow the system to learn over time
- Example: would be the interpretation of documents, such as invoices.
- There may be even less human intervention .

Socioeconomic Impact of Automation

- Automation is rapidly transforming our world, bringing both opportunities and challenges for society.

Positive Impacts:

- **Increased Productivity and Economic Growth:** Automating repetitive tasks leads to higher output, potentially boosting economic growth and living standards. (Example: Automated manufacturing lines produce goods faster, lowering costs and increasing availability.)
- **Improved Job Quality:** By taking over mundane tasks, automation can free up human workers for more engaging and creative roles. (Example: AI-powered customer service chatbots handle basic inquiries, allowing human agents to focus on complex issues.)
- **Enhanced Innovation:** Increased efficiency from automation can free up resources for research and development, fostering innovation. (Example: Automated data analysis in healthcare helps discover new treatment options.)
- **Greater Accessibility:** Automation can create new tools and services accessible to people with disabilities, promoting inclusivity. (Example: Self-driving cars offer mobility to those unable to drive traditionally.)

Dr. Divya Agarwal

Socioeconomic Impact of Automation

Negative Impacts:

- **Job Displacement:** Automation replaces some jobs, potentially leading to unemployment and decreased wages for certain groups. (Example: Selfcheckout kiosks in retail displace cashier jobs.)
- **Widening Skills Gap:** As automation demands evolve, workers may need to upskill or reskill to remain employable, potentially widening the skills gap. (Example: Manual factory workers might need to learn programming for operating automated machines.)
- **Income Inequality:** Benefits of automation might accrue disproportionately to skilled workers and owners of capital, exacerbating income inequality. (Example: Automation in finance benefits highly trained analysts but leaves lowerskilled data entry workers vulnerable.)
- **Ethical Concerns:** Issues around bias in algorithms, transparency of decisionmaking in AI systems, and potential misuse of automation require careful consideration. (Example: Algorithmic bias in hiring software could discriminate against certain groups.)

1. Operational Efficiency:

- **Streamlined Processes:** Automation optimizes workflows by eliminating manual, time-consuming tasks, reducing errors, and ensuring consistent execution. This leads to increased operational efficiency.
- **Resource Utilization:** Automated systems enable organizations to allocate resources more effectively, allowing employees to focus on strategic and creative tasks, fostering innovation.

2. Improved Customer Experience:

- **Personalization:** Automation facilitates the collection and analysis of customer data, enabling personalized interactions and services tailored to individual preferences.
- **Quick Response:** Automated customer support, chatbots, and selfservice portals enhance responsiveness, providing instant solutions and improving overall customer satisfaction.

3. Data-Driven Decision-making:

- Realtime Analytics: Automation processes generate and analyze large volumes of data in real-time, providing valuable insights for informed decision-making.
- Predictive Analytics: Automated systems leverage machine learning algorithms to predict trends, enabling organizations to proactively address challenges and capitalize on opportunities.

4. Agility and Adaptability:

- Flexibility: Automation allows organizations to quickly adapt to changing market conditions, customer demands, and regulatory requirements.
- Scalability: Automated processes can scale up or down to handle varying workloads, ensuring flexibility in resource allocation.

5. Digital Workforce:

- **Robotic Process Automation (RPA):** Integrating software robots into business processes enhances efficiency, reduces manual errors, and enables 24/7 task execution.
- **Human Automation Collaboration:** Automation complements human capabilities, fostering a collaborative environment where humans focus on complex tasks, creativity, and decision-making.

6. Cost Savings:

- **Reduced Operational Costs:** Automation reduces the need for extensive manual labor, leading to cost savings in terms of salaries, training, and employee turnover.
- **Resource Optimization:** Automated systems optimize resource utilization, reducing waste and inefficiencies, further contributing to cost reduction.

7. Digital Integration:

- **Connected Ecosystems:** Automation facilitates the integration of diverse digital systems, creating a seamless and interconnected ecosystem.
- **APIs and Integration Platforms:** Automated processes often rely on APIs and integration platforms to connect different software applications, promoting interoperability.

8. Compliance and Risk Management:

- **Consistent Compliance:** Automation ensures adherence to regulatory requirements by enforcing standardized processes and reducing the risk of human error.
- **Audit Trails:** Automated systems generate detailed audit trails, aiding in compliance audits and risk management assessments.

9. Innovation Acceleration:

- **Focus on Creativity:** By automating routine tasks, employees can allocate more time to innovative and strategic activities, fostering a culture of continuous improvement.
- **Experimentation:** Automation allows organizations to experiment with new technologies and approaches, accelerating the pace of digital innovation.

Role of Automation in Business

- **Efficiency and Productivity:**
 - **Automating Repetitive Tasks:** RPA is designed to handle rule-based, repetitive tasks that are time-consuming for humans. By automating these tasks, businesses can significantly improve operational efficiency and productivity.
 - **24/7 Operations:** RPA bots can work around the clock without breaks, leading to continuous operations and faster task completion.
- **Cost Reduction:**
 - **Labor Cost Savings:** RPA reduces the need for manual labor in routine tasks, resulting in significant cost savings in terms of salaries, training, and employee benefits.
 - **Error Reduction:** Automation minimizes errors in data entry and processing, preventing costly mistakes that may arise from manual work.
- **Accuracy and Quality:**
 - **Precision and Consistency:** RPA ensures a high level of accuracy and consistency in task execution, reducing the risk of human errors.
 - **Quality Assurance:** Automated processes can follow predefined rules and quality standards, contributing to improved overall product and service quality.

Role of Automation in Business

- **Scalability:**
 - **Handling Increased Workloads:** RPA systems can easily scale to handle increased workloads without a proportional increase in human resources, providing scalability to meet business demands.
 - **Adapting to Business Growth:** As businesses expand, RPA enables seamless scaling of operations to accommodate growth.
- **Enhanced Customer Experience:**
 - **Faster Response Times:** RPA can automate customer interactions, leading to quicker response times and improved customer satisfaction.
 - **Personalization:** By automating data analysis, businesses can offer personalized services and experiences to their customers.
- **Data Accuracy and Management:**
 - **Data Integration:** RPA can integrate with various systems, ensuring smooth data flow between different applications and databases.
 - **Data Validation:** Automated processes validate and cleanse data, contributing to the maintenance of accurate and reliable data repositories.

Role of Automation in Business

- **Employee Empowerment:**
 - **Focus on Value-Added Tasks:** RPA allows employees to focus on higher-value tasks that require creativity, critical thinking, and decision-making, enhancing job satisfaction.
 - **Skill Enhancement:** Employees can acquire and enhance skills related to managing and overseeing automated processes, contributing to professional development.
- **Agility and Adaptability:**
 - **Quick Implementation:** RPA can be implemented relatively quickly, allowing businesses to adapt to changing market conditions and demands.
 - **Process Modifications:** Automated processes can be easily modified and updated to accommodate changes in business requirements.
- **Innovation and Strategic Focus:**
 - **Innovation Acceleration:** With routine tasks automated, businesses can allocate more resources to innovation and strategic initiatives, driving continuous improvement.
 - **Strategic Decision-Making:** RPA provides accurate and timely data, supporting strategic decision-making by business leaders.

Role of Automation in Business

- **Compliance and Risk Management:**
 - **Consistent Compliance:** RPA ensures that processes adhere to regulatory requirements, reducing the risk of non-compliance and associated penalties.
 - **Audit Trails:** Automated processes generate detailed audit trails, providing transparency and supporting risk management efforts.

- RPA vendor market has been showing continual and steady growth.
- While the largest market is the US, followed by the UK, the market in Asia Pacific Countries(APAC) is also showing considerable progress.
- Successful pilot projects and increased customer satisfaction among the early adopters of RPA will encourage new players to adopt this technology.
- There is growing demand for RPA, especially in industries that need large scale deployments.
- The major markets for RPA are banking and finance, healthcare and pharmaceuticals, telecom and media, and retail.

1.Automation Anywhere

- Automation Anywhere helps to automate business processes for companies. They focus on RPA, cognitive data (machine learning and natural language processing), and business analytics. Their bots are capable of handling both structured as well as unstructured data.
- The system has three basic components:
 1. *A development client* for the creation of a bot
 2. *A runtime environment* for the deployment of a bot
 3. *A centralized command system* for handling multiple bots, analyzing their performance:

2. UiPath

- UiPath is an RPA technology vendor who designs and delivers software that helps automate businesses.
- The RPA platform consists of three parts:
 - -> *UiPath Studio* to design the processes
 - -> *UiPath Robot* to automate tasks designed in UiPath Studio
 - -> *UiPath Orchestrator* to run and manage the processes

3. Blue Prism

- Blue Prism aims to provide automation that enterprises can use according to their needs.
- Blue Prism aims to do this by providing automation that is scalable, configurable, and centrally managed.
- It sells its software through its partners, some of which are Accenture, Capgemini, Deloitte, Digital Workforce Nordic, HPE, HCL, IBM, TCS, Tech Mahindra, Thoughtonomy, and Wipro

4. Work Fusion

- Work Fusion offers automation that is based on RPA and machine learning.
- It delivers software as a solution for automating high volume data.
- Work Fusion enables man and machine to work in tandem while managing, optimizing, or automating tasks.

5. Thoughtonomy and 6. KOFAX

Thoughtonomy

- Thoughtonomy delivers software that helps automate business and IT processes.
- It uses Blue Prism and other automation software and customizes it

6. KOFAX

- Kofax's RPA platform is capable of automating and delivering processes that are repetitive and rule-based.
- It uses Robots for extracting and consolidating information.
- The software platform consists of a management console to deploy and manage bots, Robot performance, and a monitoring system.
- This software can also group together high priority tasks that should be completed first by the Robot during times of high workload. Kofax's software, however, doesn't have machine learning.

UiPath Studio is a key component of the UiPath business automation platform. It is an Integrated Development Environment (IDE) that allows automation developers to create, design, and manage automation workflows.

It provides a user-friendly, drag-and-drop interface for building automation processes without the need for extensive coding. UiPath Studio integrates with various applications and systems, allowing users to automate processes across different platforms. It supports integration with APIs, databases, web services, and more.

- UiPath Studio is a powerful tool that encompasses many features and capabilities that enable you to build the best automation solutions. To use Studio at its full potential, you must be well acquainted with the user interface.
- The Studio user interface (UI) consists of:
 - The **Backstage View**
 - The **Ribbon**
 - The **Command Palette**
 - The **Panels**
 - The **Status bar**

- Studio comes with two types of profiles for developers and business users:
- **Studio** offers a plethora of tools for designing complex attended, unattended, and testing automation. Studio also supports both low-code and coded automation:
 - **Low-code automation** use a visual interface with drag-and-drop activities, making it accessible for a broad user base.
 - **Coded automation** allow you to create automation using code, which is ideal for more complex and collaborative projects. These automations use UiPath services (which are the equivalent of activity packages), APIs (similar to activities), external .NET NuGet packages, and your custom C# classes within Studio.
- **StudioX** enables business users with little or no coding experience to automate their tasks. Check out the [StudioX](#) guide.

UiPath Studio



Collaboration

Git, TFS/VSTS, SVN Integration
File Diff and Compare Changes



Usability

Command Palette
Keyboard Shortcuts
Global and Project Settings



Productivity

Manage Dependencies
Activity Project Settings
Selectors



Reusability

Libraries
Custom Templates

UiPath Studio



Code Quality

Workflow Analyzer
Create Custom Rules
Validate



Debugging

Breakpoints
Call Stack, Immediate, Locals, Watch
Test Activity and Create Test Bench



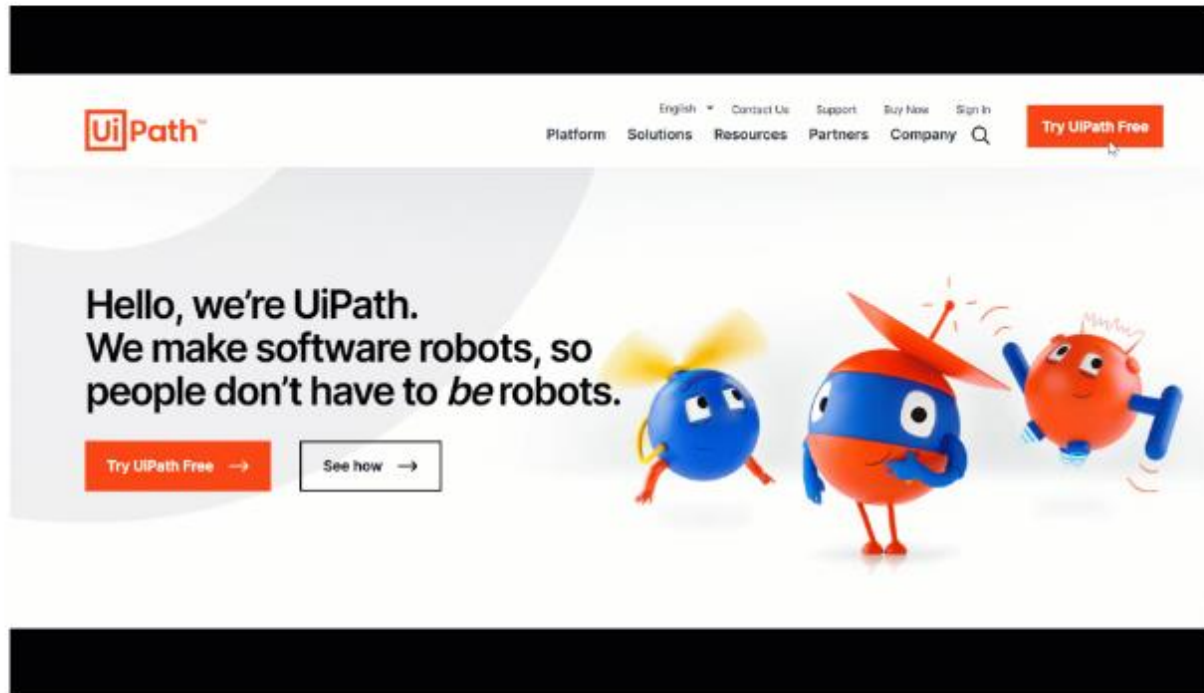
Extensibility

Project Templates
Browser Extensions
Automate SAP and Citrix Technologies

Steps to Install UiPath StudioX

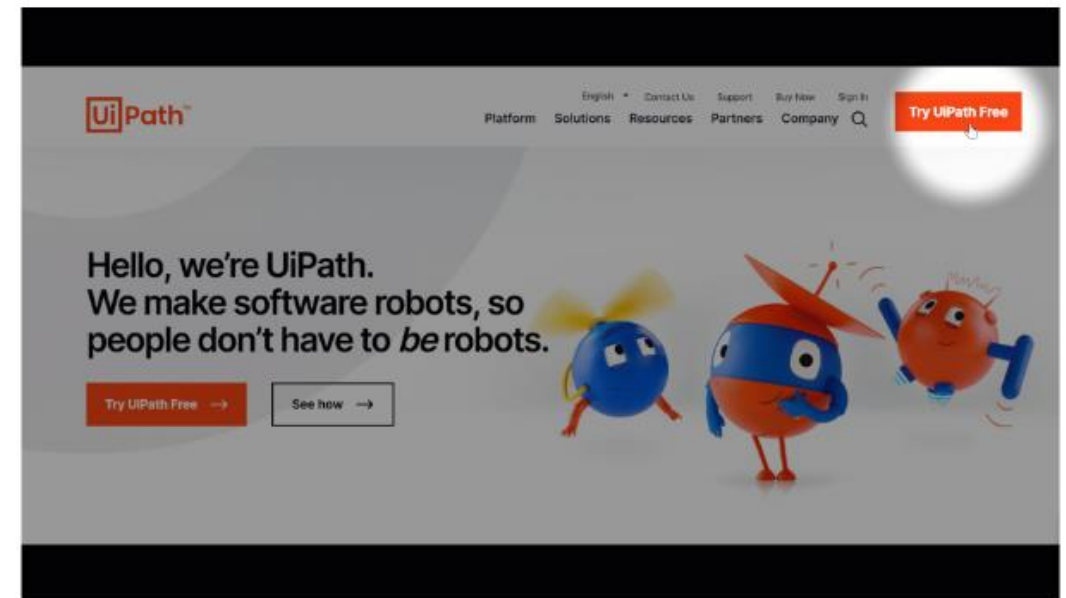
Step 1

Create your Community Cloud platform account



Step 1

Create your Community Cloud platform account



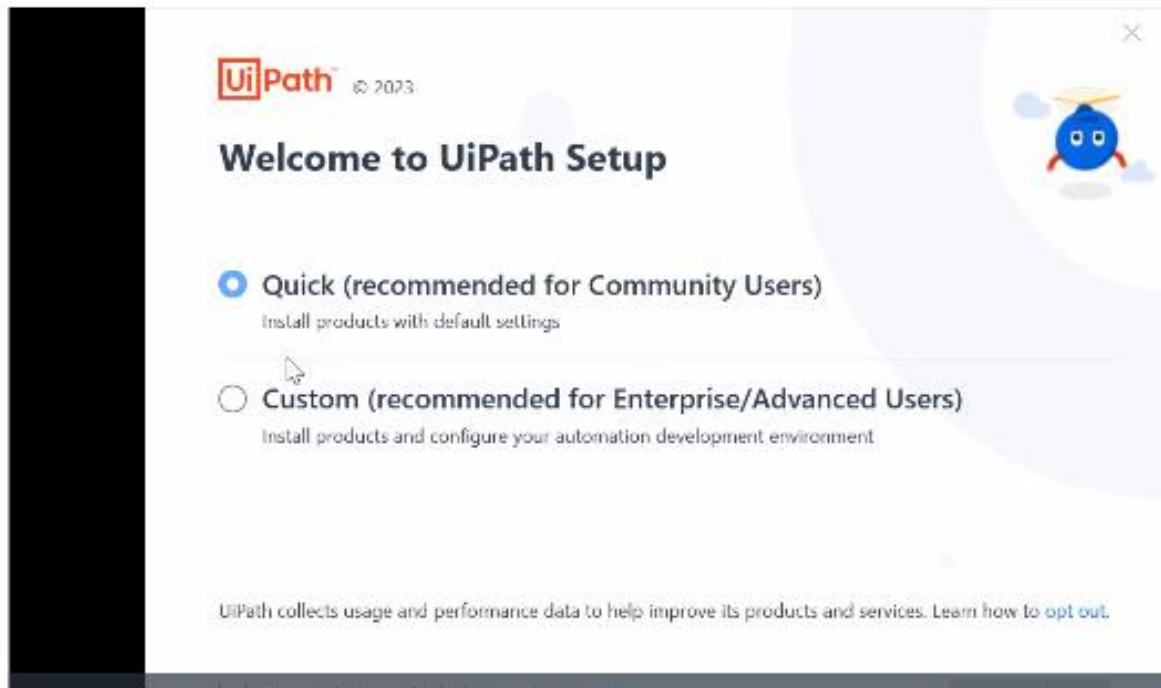
Steps to Install UiPath StudioX

Step 2: Click on
download Studio
option

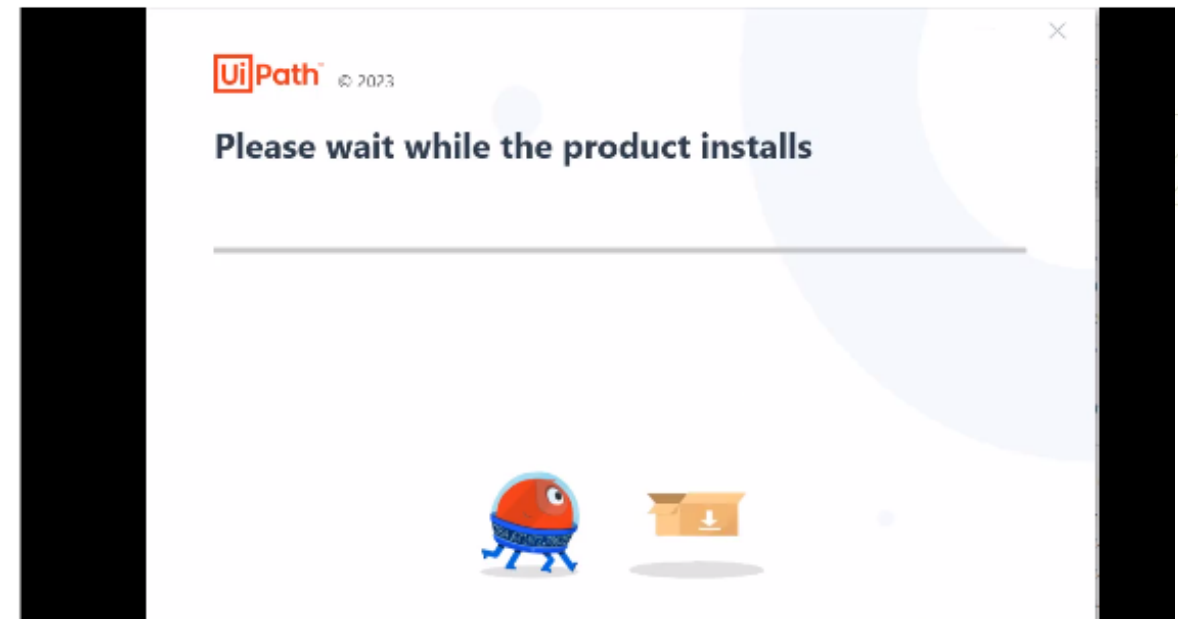
The screenshot shows the UiPath Automation Cloud portal interface. The browser tabs include 'Home - Automation Cloud', 'UiPath Academy', and 'UiPath Academy'. The address bar shows 'cloud.uipath.com/usarggsipudelhi/portal_/home'. The portal header displays the 'UiPath Automation Cloud' logo and a user profile icon. The main content area greets the user 'Good morning, Divya Agarwal' and features a banner for 'Redefining the customer experience' with a 'Join the summit' button. Below this, there are sections for 'Orchestrator Services' (showing 'DefaultTenant' updated 369 days ago) and 'Start from popular templates'. On the right sidebar, there is a 'Boost your productivity effortlessly' section with a 'Download Studio' button, and a 'What's new' section with links for 'View listing', 'See the documentation', and 'Get the report'. The Windows taskbar at the bottom shows the search bar, task view, and various application icons, with the system clock indicating 09:32 on 06-02-2024.

Steps to Install UiPath StudioX

Install UiPath Studio



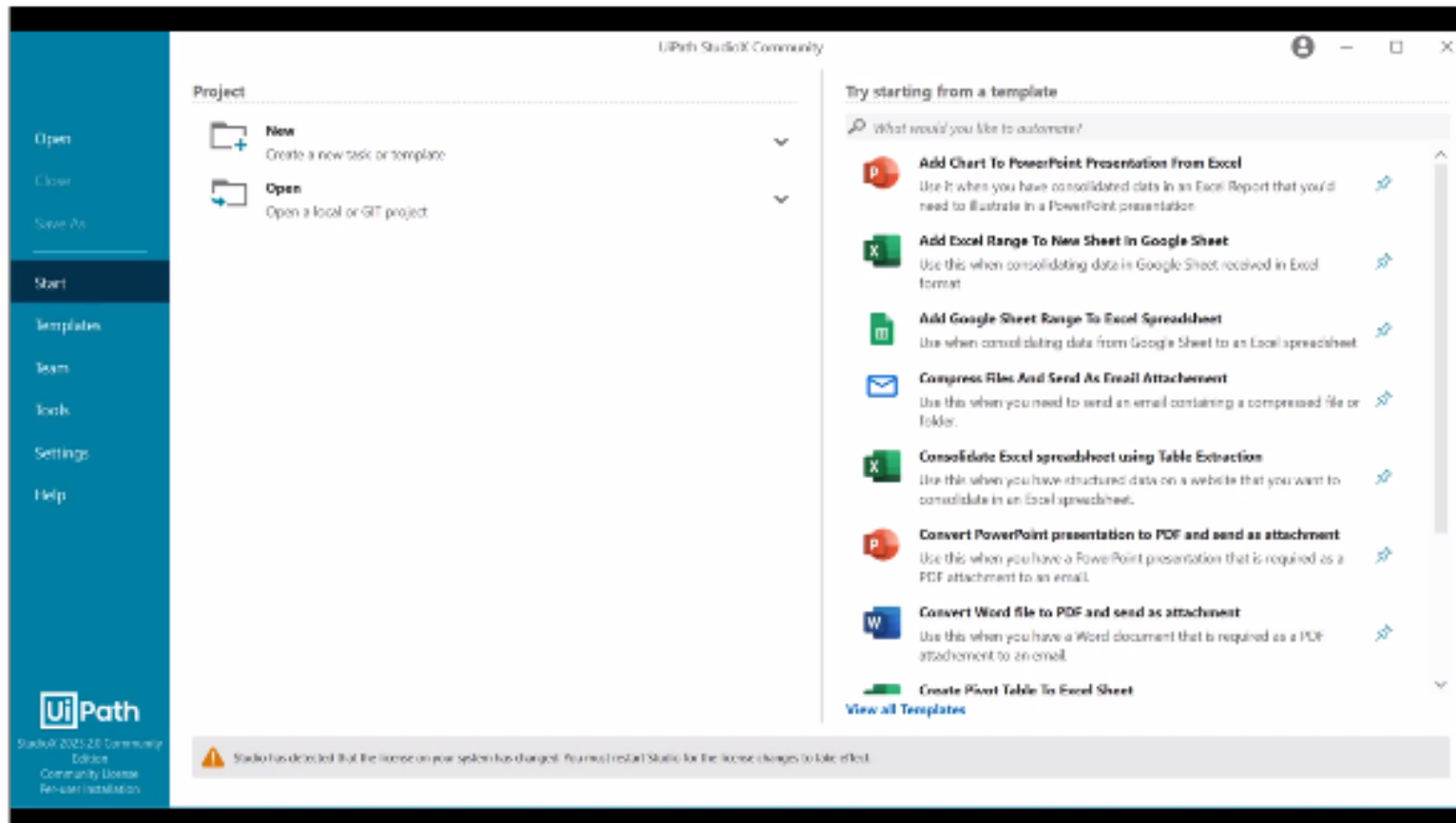
Install UiPath Studio



Step 3: Install Ui Path Studio

Steps to Install UiPath StudioX

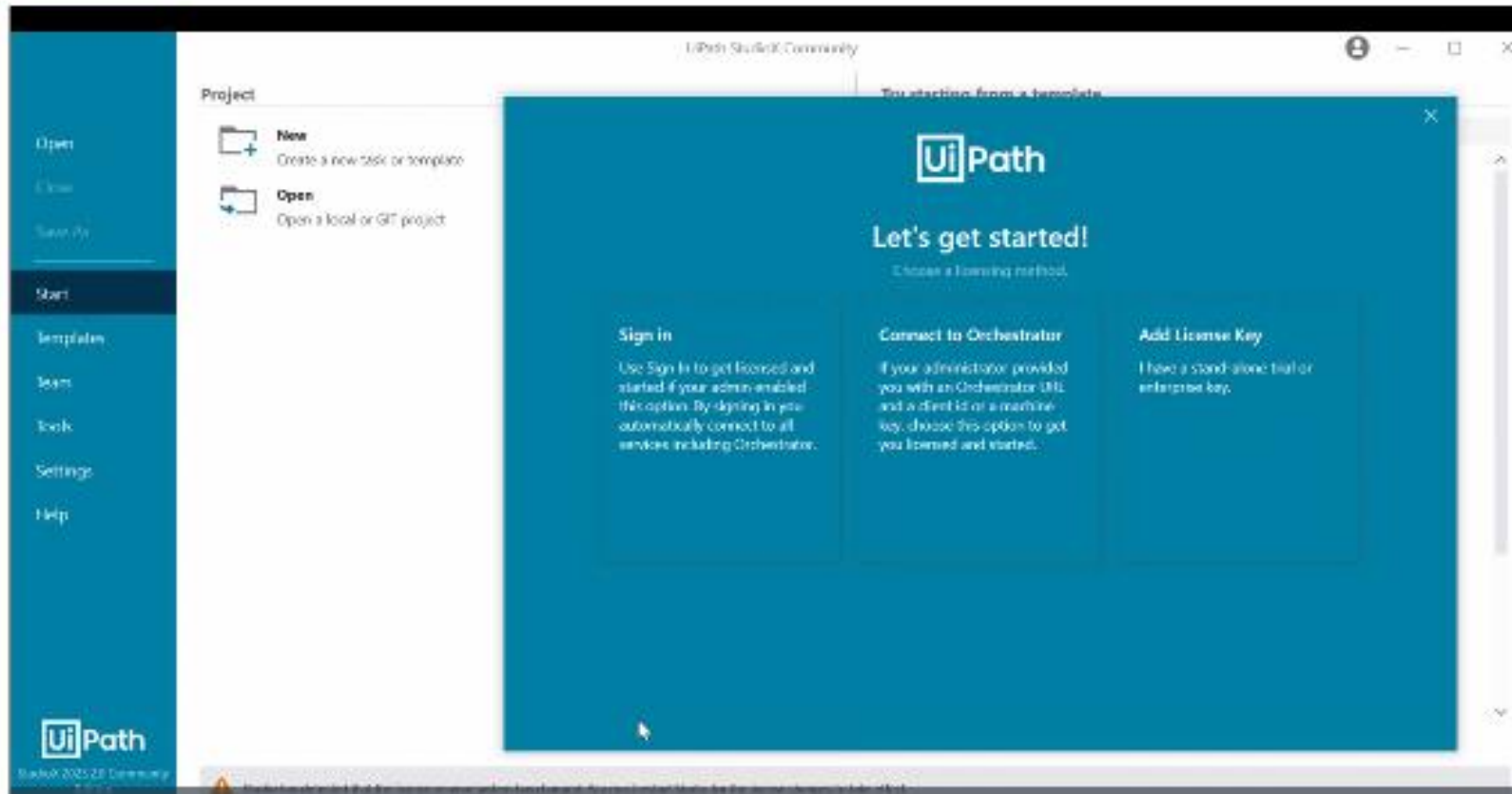
Signing into you Automation Cloud account



Step 4: Signing into you Automation Cloud account

Steps to Install UiPath StudioX

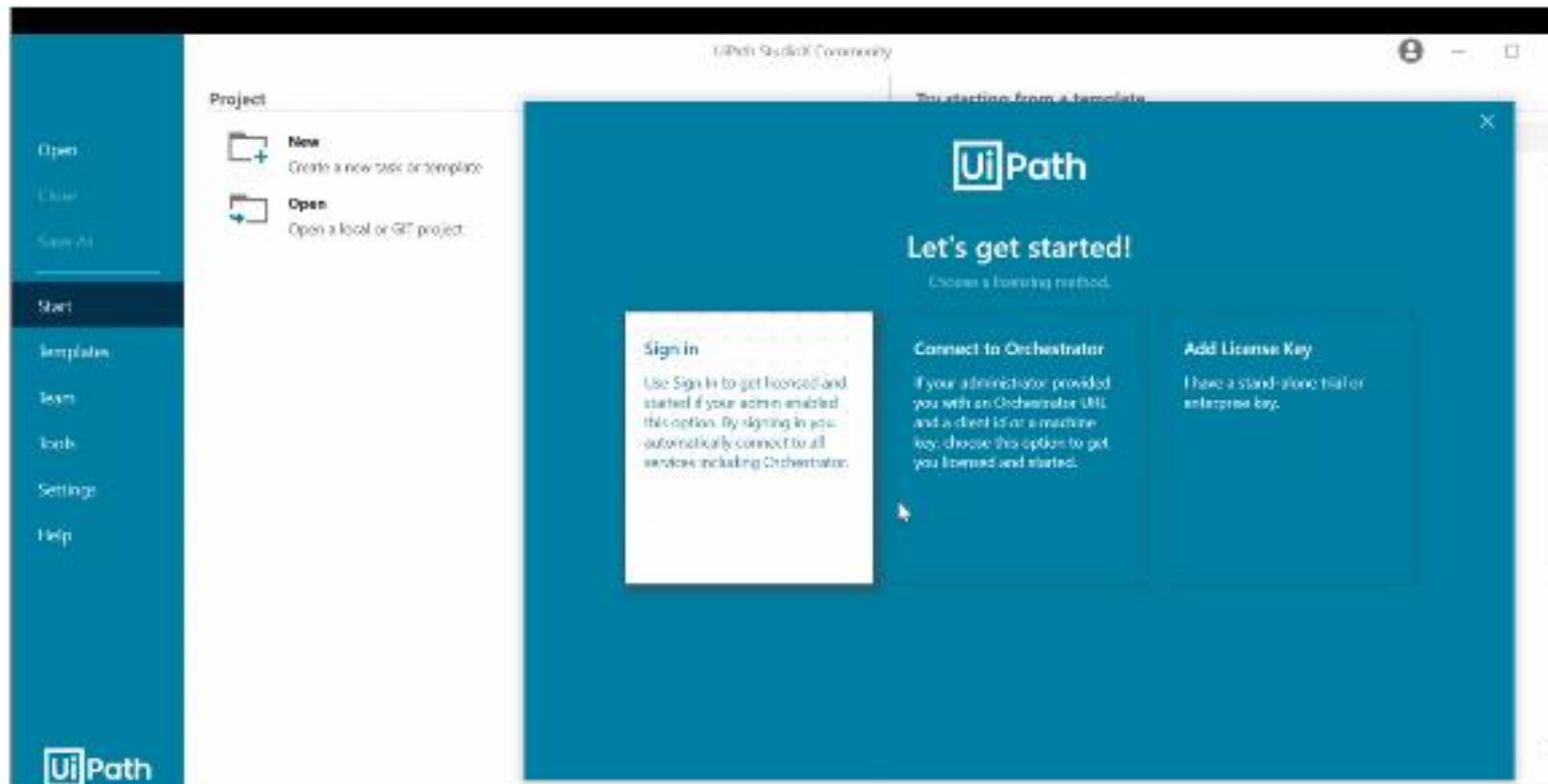
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Step 4: Signing into you Automation Cloud account

Steps to Install UiPath StudioX

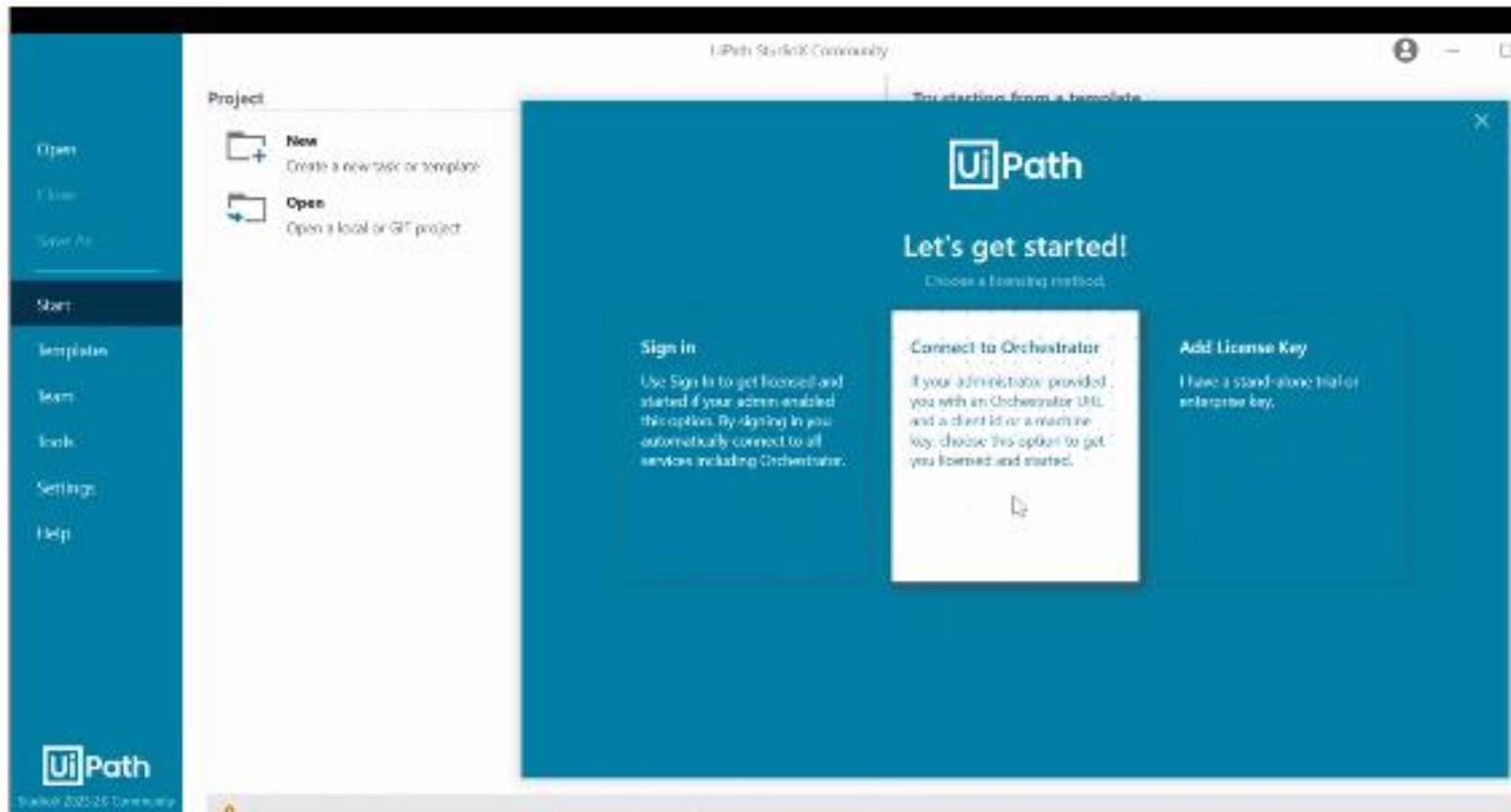
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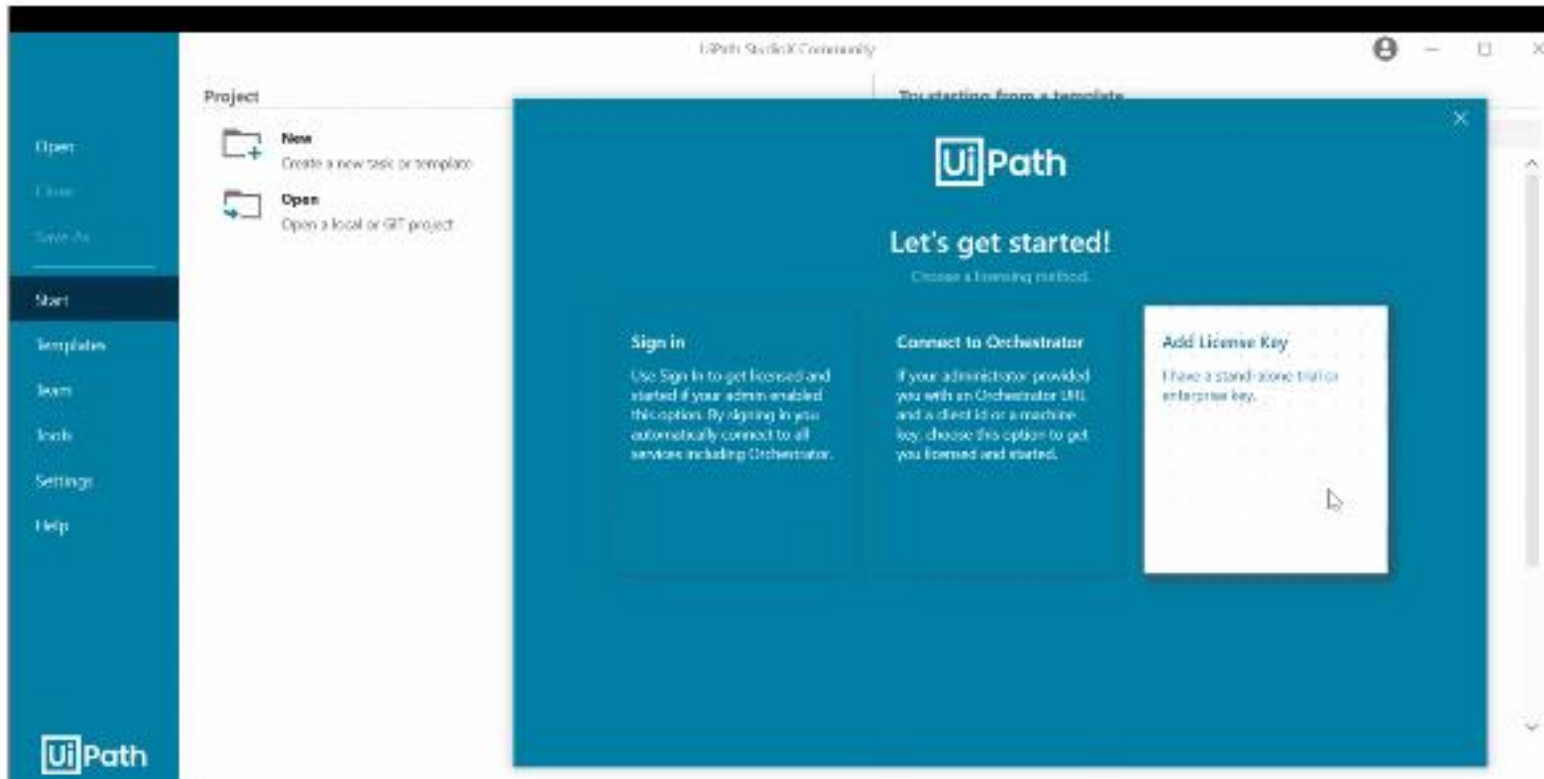
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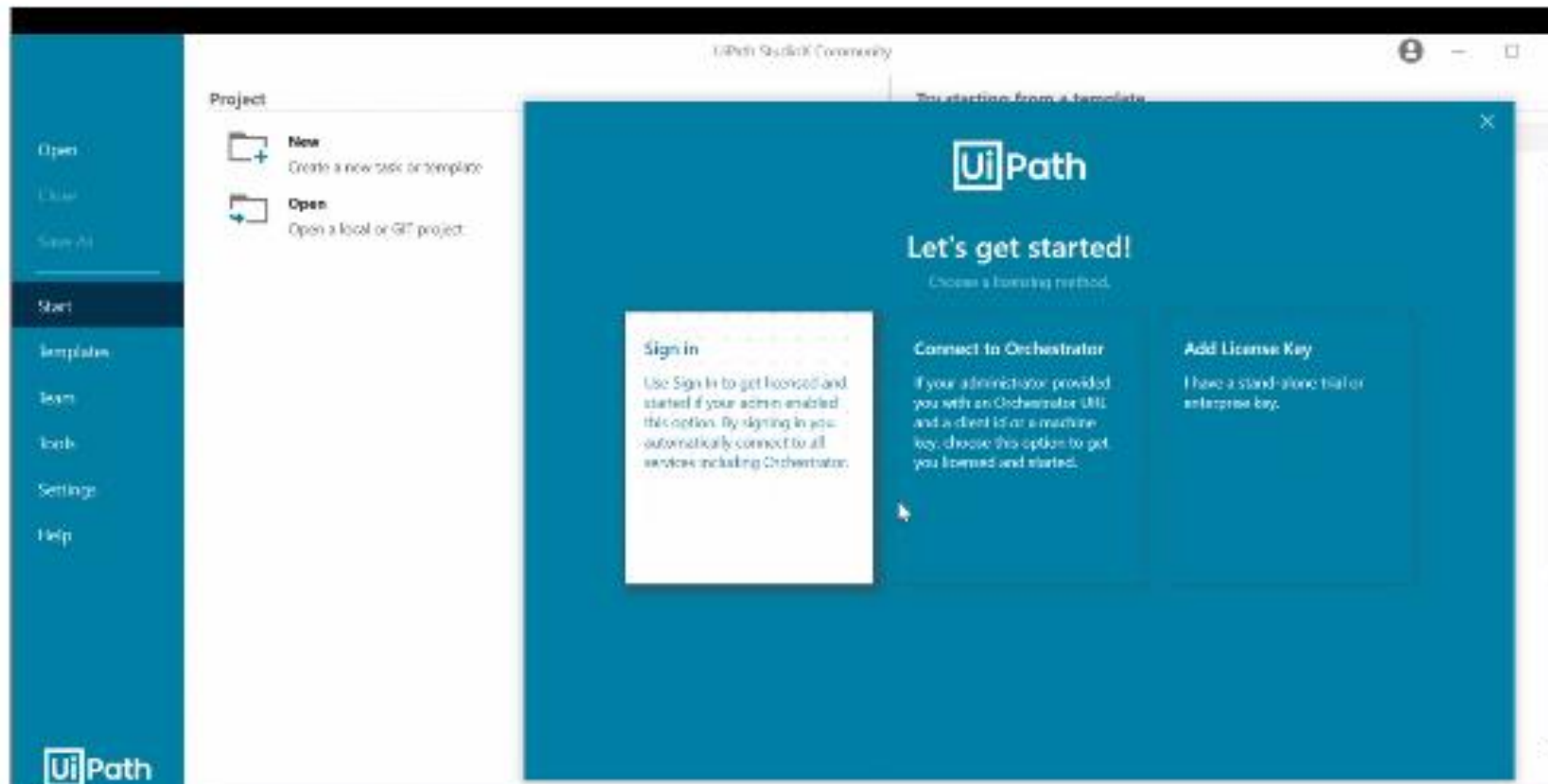
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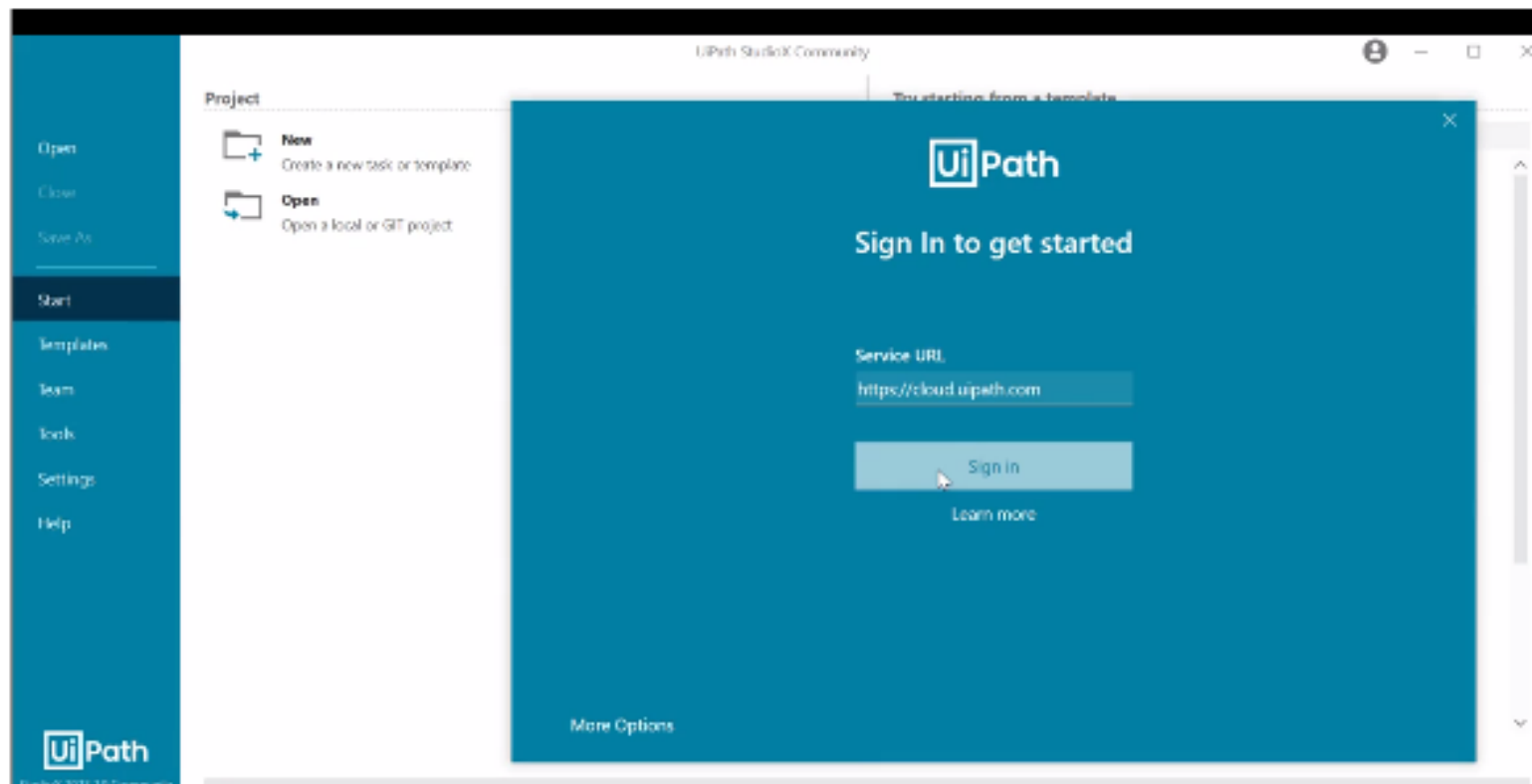
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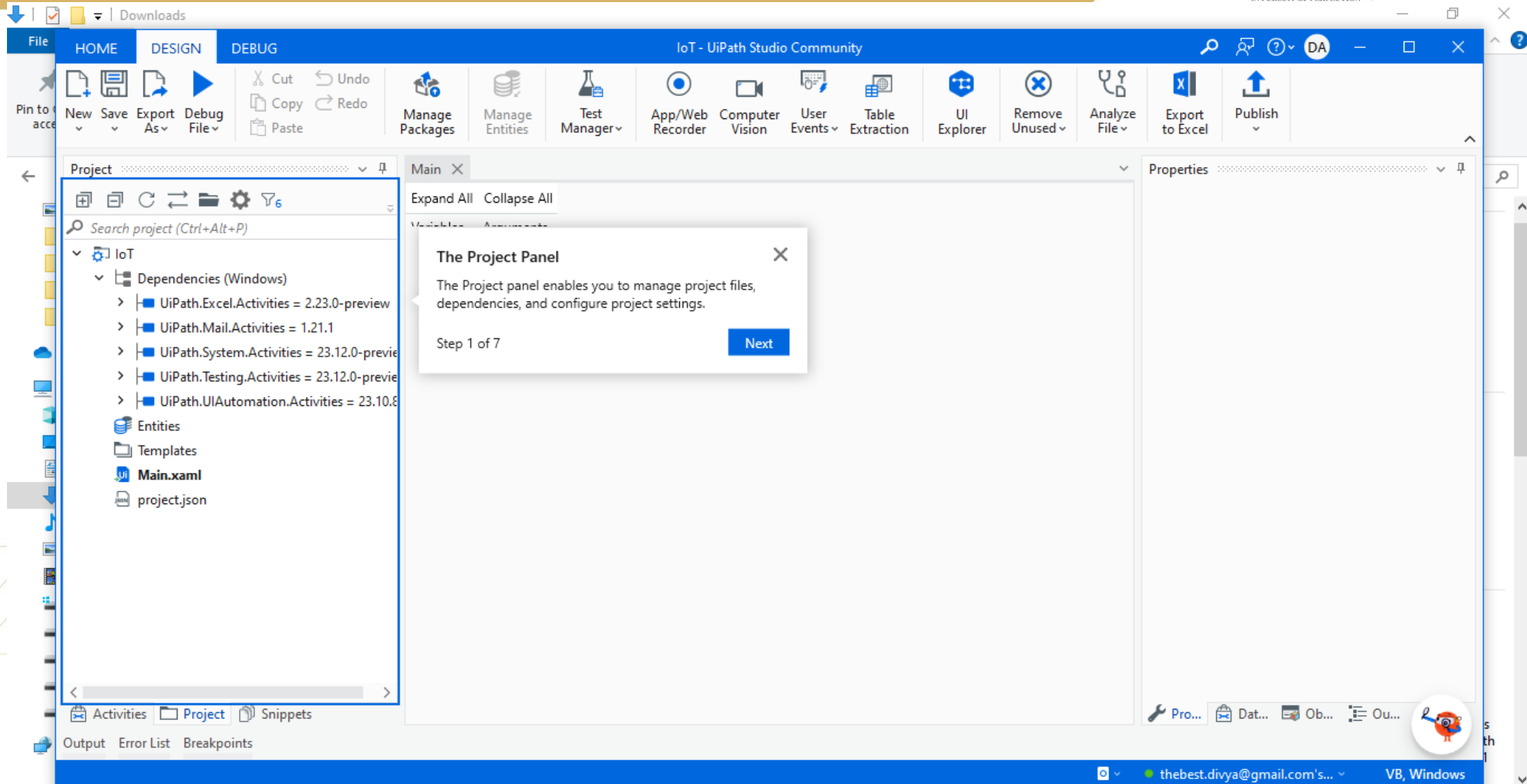
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Signing into your Automation Cloud account

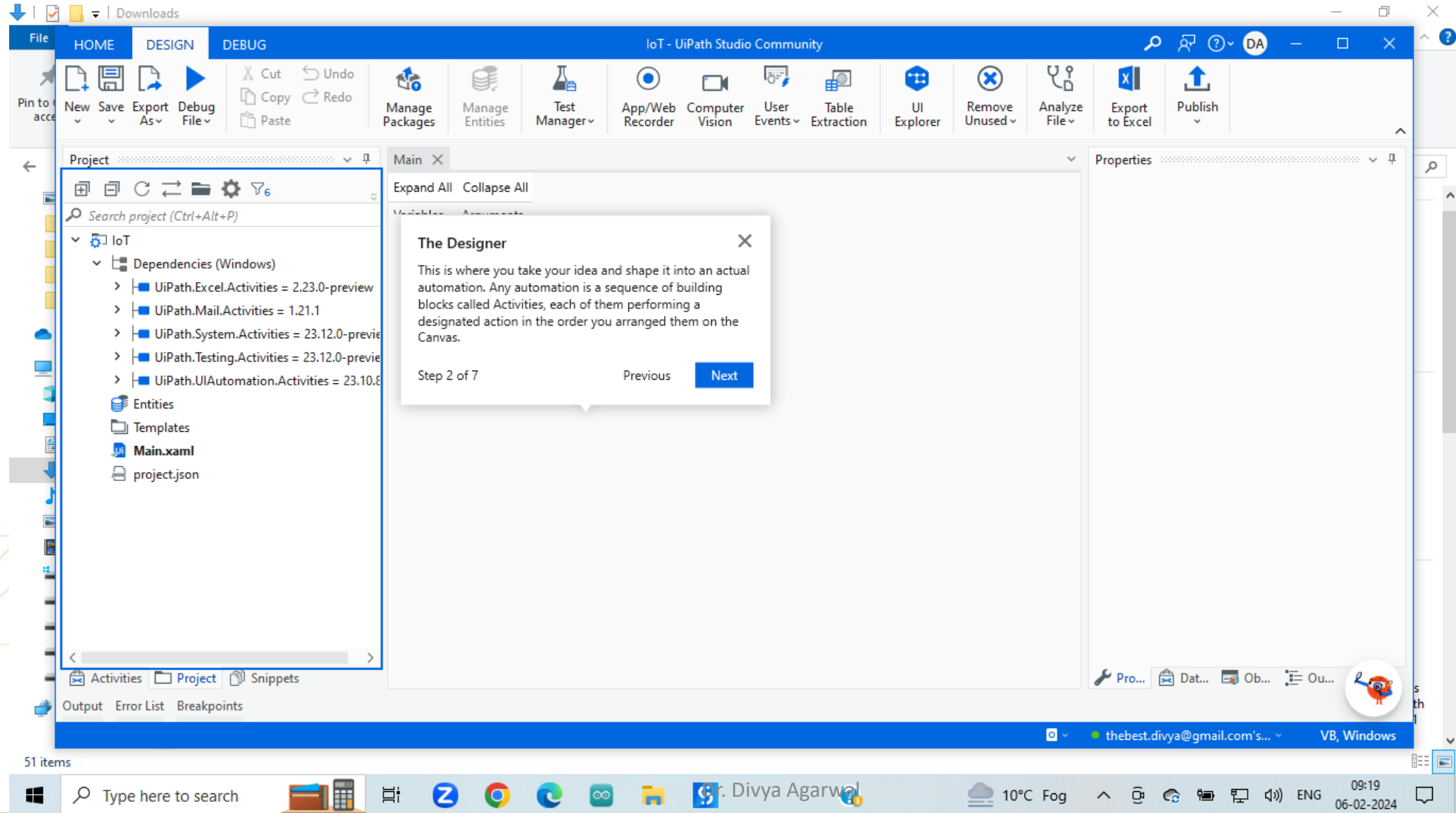


Step 4: Signing into your Automation Cloud account

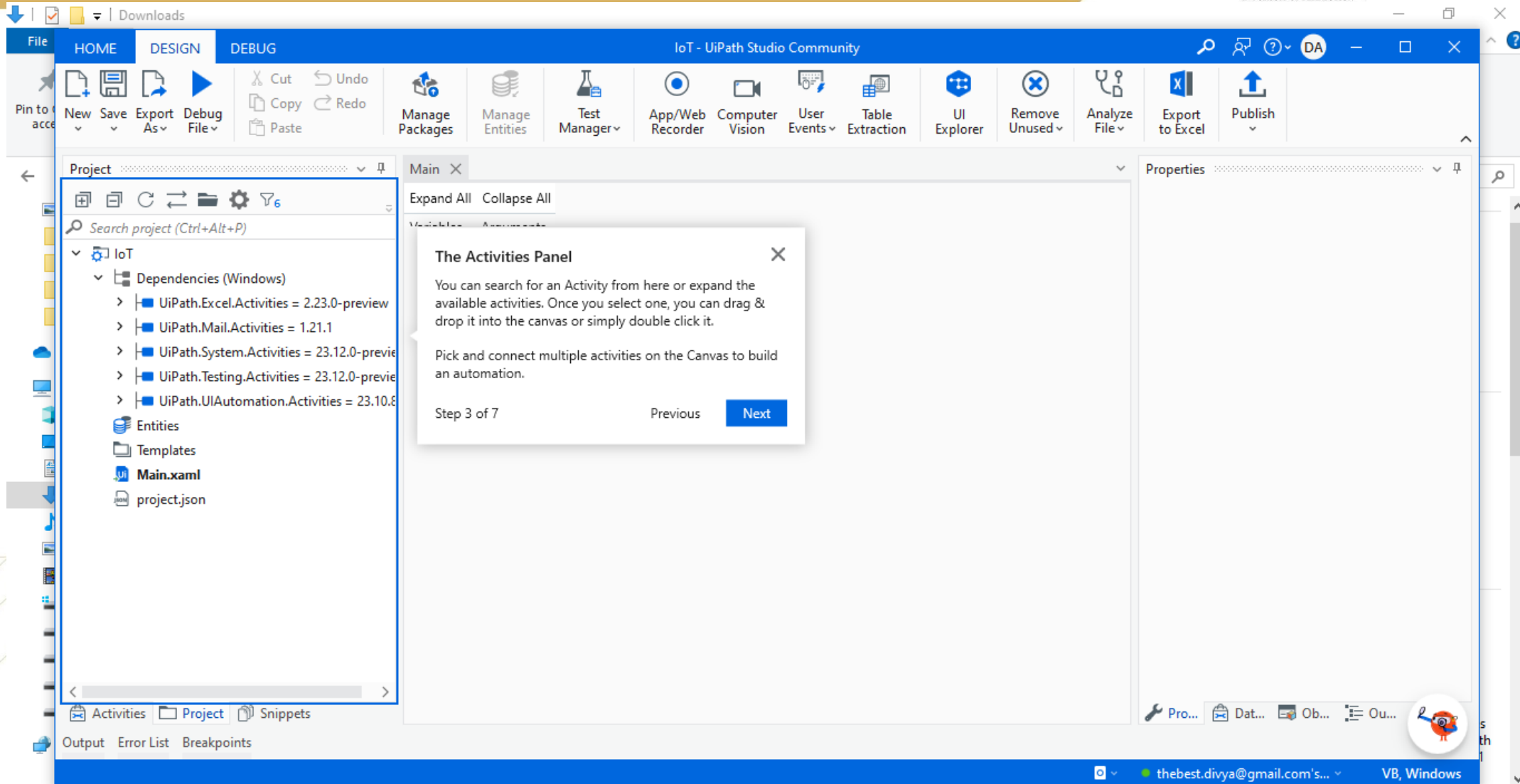
Introduction to StudioX IDE



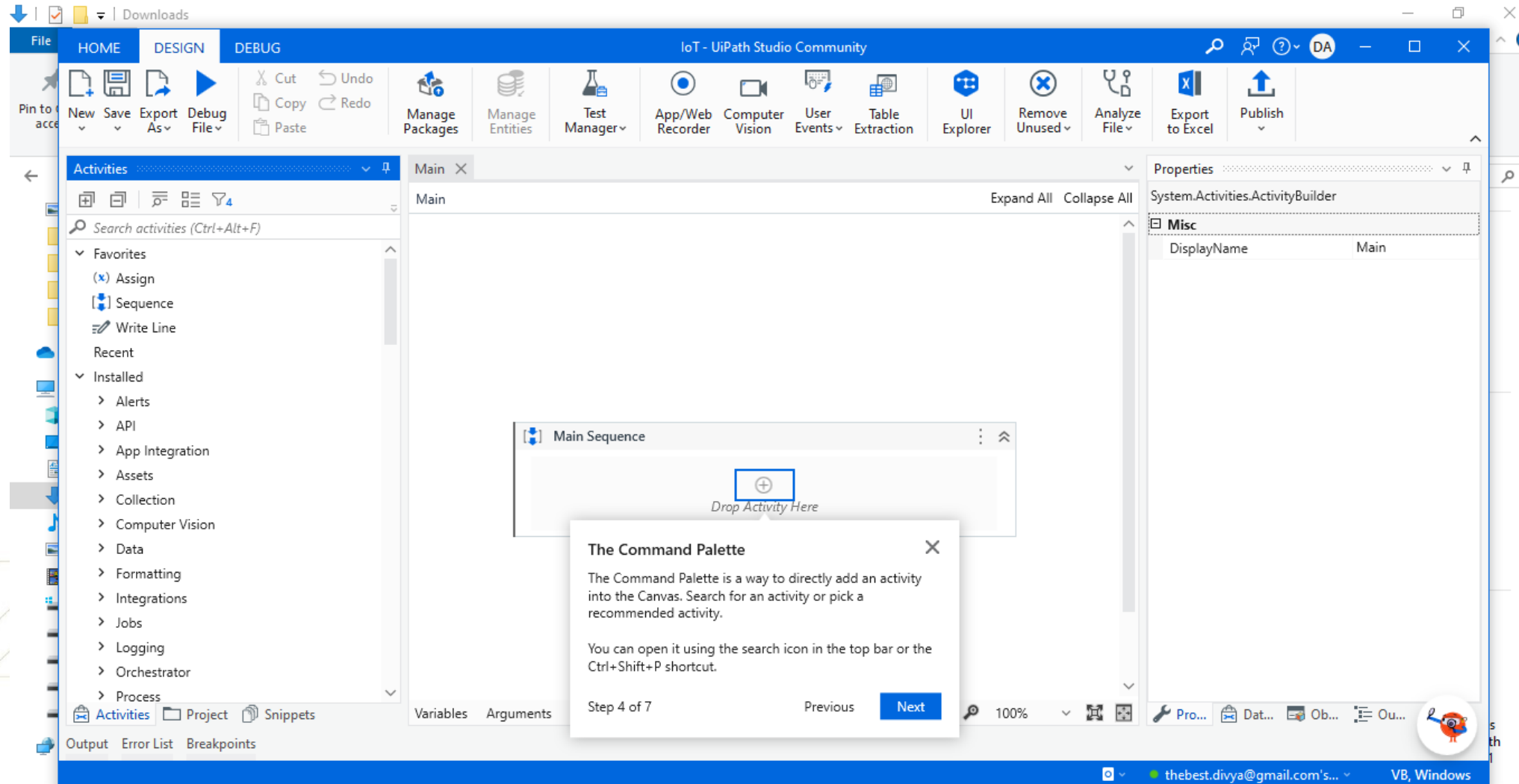
Introduction to StudioX IDE



Introduction to StudioX IDE



Introduction to StudioX IDE



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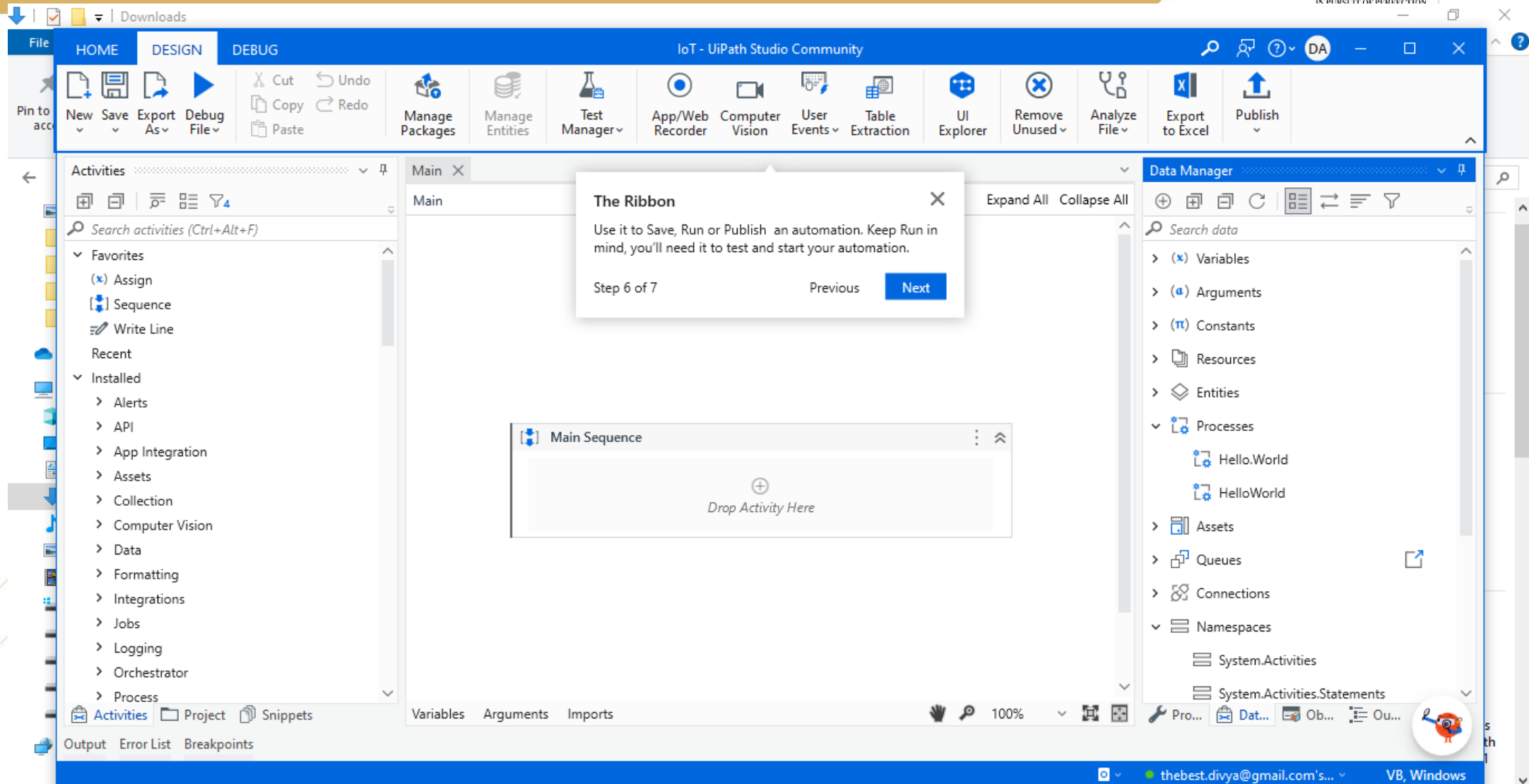
10°C Fog

09:19
06-02-2024

Introduction to StudioX IDE

The screenshot displays the StudioX IDE interface, which is divided into several panes. The top pane is the 'File' menu, which includes options like 'New', 'Save', 'Export As', 'Debug File', 'Cut', 'Copy', 'Paste', 'Undo', 'Redo', 'Manage Packages', 'Manage Entities', 'Test Manager', 'App/Web Recorder', 'Computer Vision', 'User Events', 'Table Extraction', 'UI Explorer', 'Remove Unused', 'Analyze File', 'Export to Excel', and 'Publish'. Below the 'File' menu is the 'Activities' pane, which contains a search bar and a list of activities categorized into 'Favorites' (Assign, Sequence, Write Line) and 'Installed' (Alerts, API, App Integration, Assets, Collection, Computer Vision, Data, Formatting, Integrations, Jobs, Logging, Orchestrator, Process). The main workspace is the 'Main Sequence' editor, which shows a 'Main' sequence with a 'Drop Activity Here' prompt. A 'Data Manager' window is open in the foreground, displaying a list of data assets: Variables, Arguments, Constants, Resources, Entities, Processes (Hello.World, HelloWorld), Assets, Queues, Connections, and Namespaces (System.Activities, System.Activities.Statements). The 'Data Manager' window also includes a search bar and a 'Search data' button. The bottom status bar shows the user's email address 'thebest.divya@gmail.com's...' and the current window title 'VB, Windows'. The Windows taskbar at the very bottom shows the search bar, task view, and several open applications, including a calculator, a file explorer, and a web browser. The system tray on the right shows the date and time as '09:20 06-02-2024' and the temperature as '10°C Fog'.

Introduction to StudioX IDE



Introduction to StudioX IDE

The screenshot displays the UiPath StudioX IDE interface. The top menu bar includes 'File', 'HOME', 'DESIGN', and 'DEBUG'. The 'HOME' tab is active, showing a 'The Home tab' tooltip that explains its purpose: 'Home is where you can manage your automations, create new ones, browse templates or even find learning resources - consider it your backstage.' The tooltip also indicates 'Step 7 of 7' and provides 'Previous' and 'Got it' buttons.

The left sidebar contains a 'Favorites' section with items like 'Assign', 'Sequence', and 'Write Line', and an 'Installed' section with various activity categories like 'Alerts', 'API', 'App Integration', 'Assets', 'Collection', 'Computer Vision', 'Data', 'Formatting', 'Integrations', 'Jobs', 'Logging', 'Orchestrator', and 'Process'. Below this are tabs for 'Activities', 'Project', and 'Snippets'.

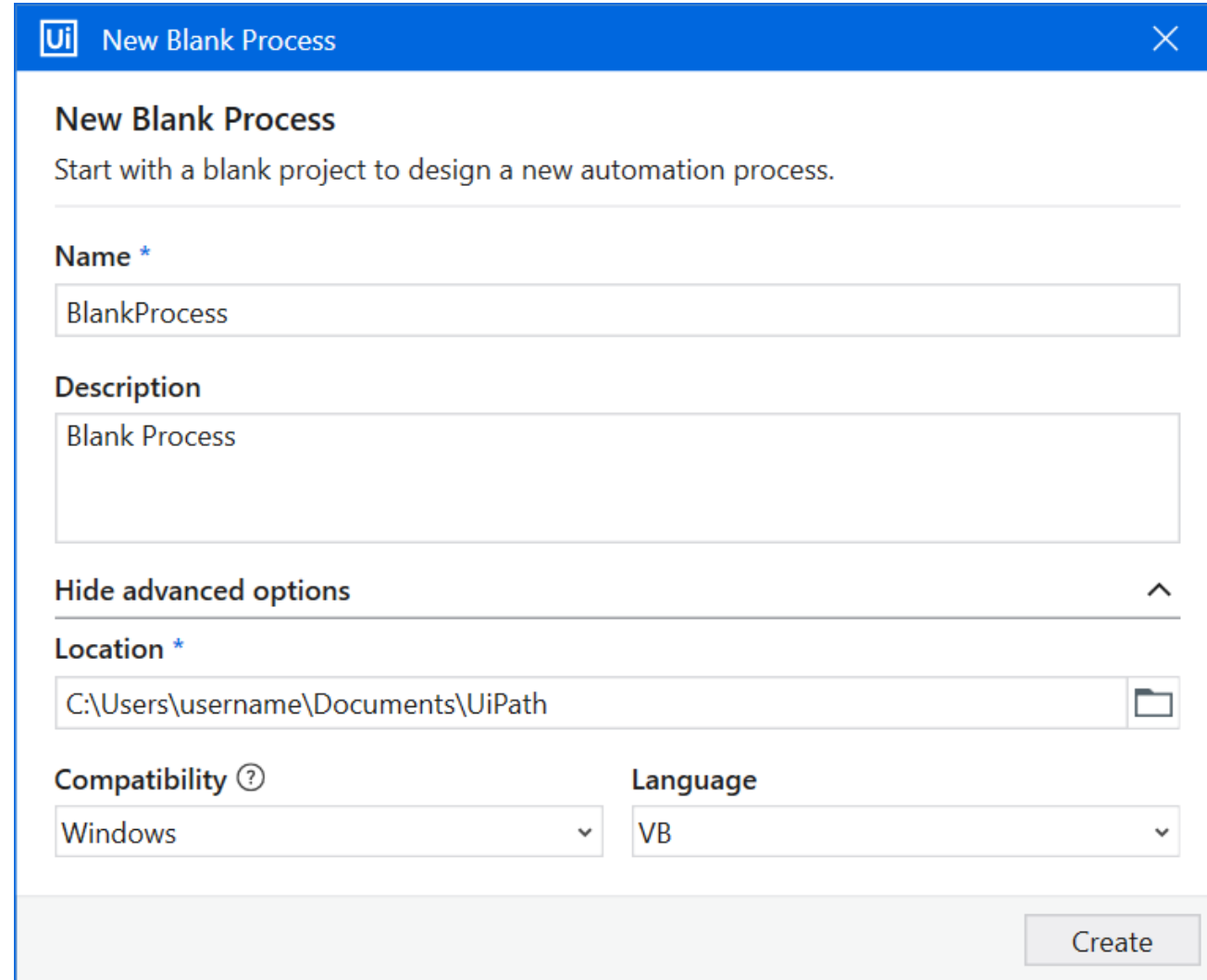
The main workspace shows a 'Main Sequence' editor with a 'Drop Activity Here' prompt. The right sidebar features the 'Data Manager' panel, which lists various data types: Variables, Arguments, Constants, Resources, Entities, Processes (including 'Hello.World' and 'HelloWorld'), Assets, Queues, Connections, and Namespaces (including 'System.Activities' and 'System.Activities.Statements').

The bottom status bar shows the user's email 'thebest.divya@gmail.com's...' and the operating system 'VB, Windows'. The Windows taskbar at the very bottom includes a search bar, task view, and several application icons, along with system tray information like '10°C Fog' and the date '06-02-2024'.

Creating a Basic Process

- <https://www.youtube.com/watch?v=OyQAhrCBr9U>

Step 1: Launch Studio. In the HOME Backstage view, click **Process** to create a new project. The **New Blank Process** window is displayed



Ui New Blank Process

New Blank Process
Start with a blank project to design a new automation process.

Name *
BlankProcess

Description
Blank Process

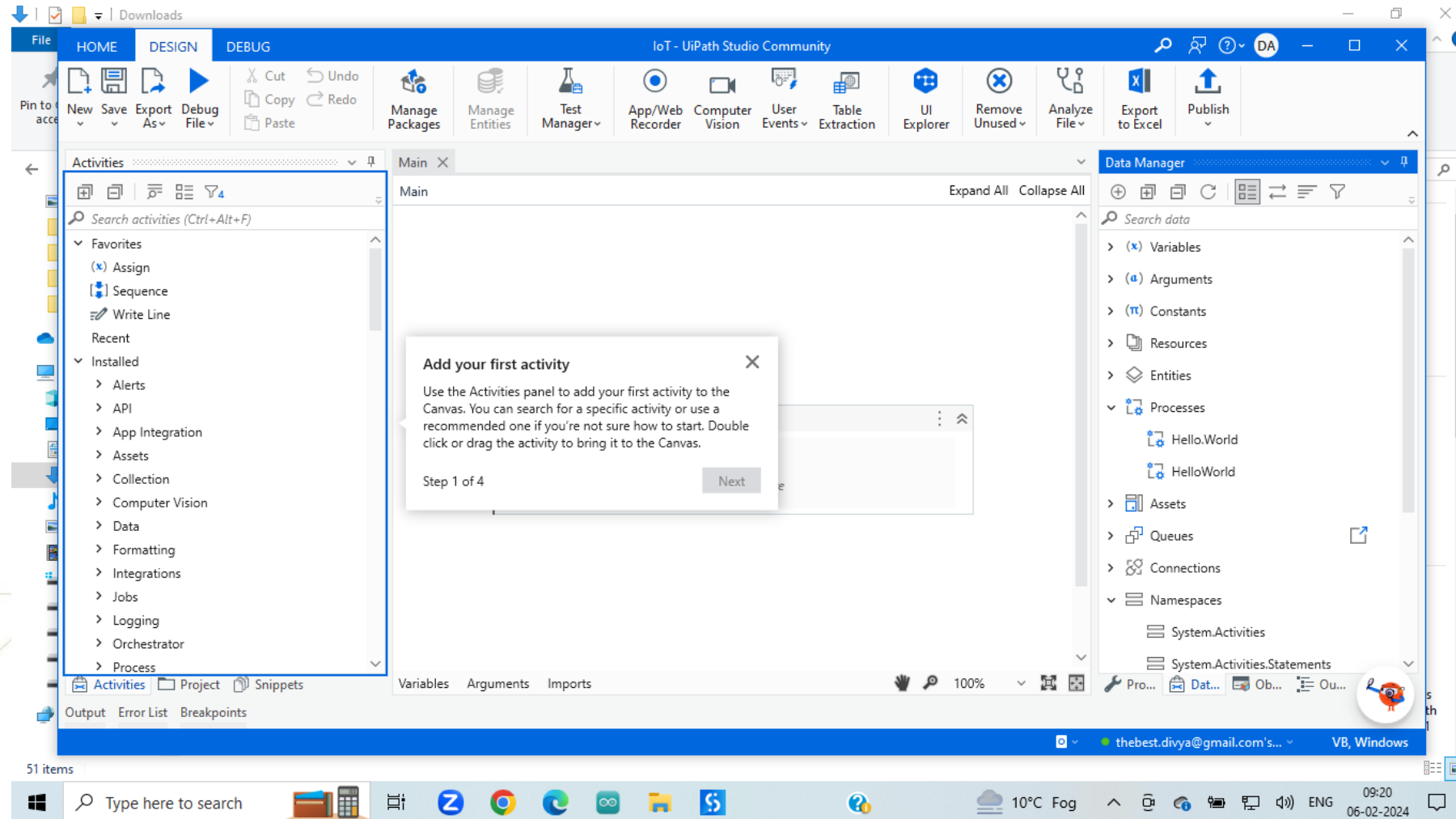
Hide advanced options ^

Location *
C:\Users\username\Documents\UiPath

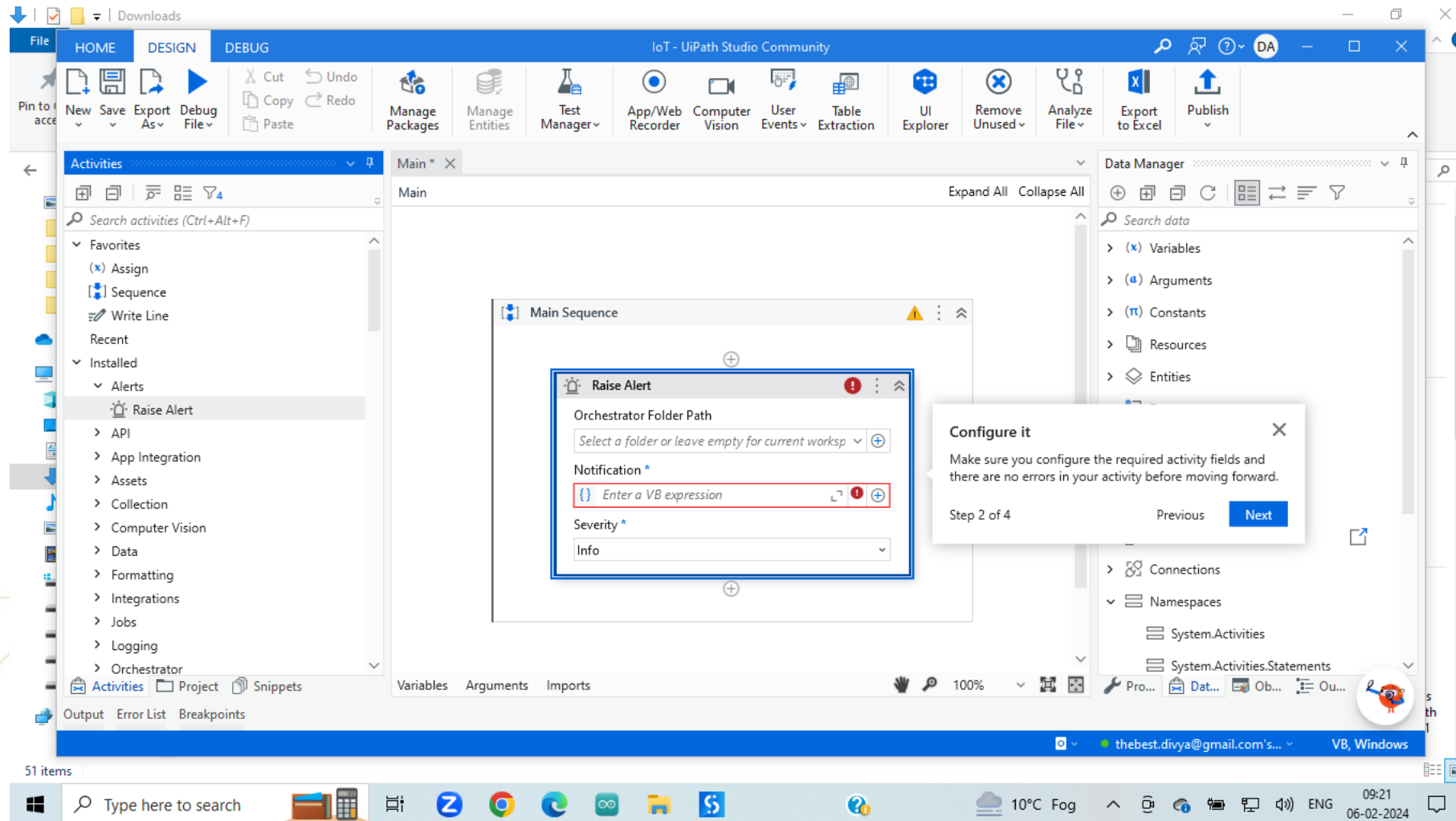
Compatibility ? Windows ▾ **Language** VB ▾

Create

Steps to create a process



Steps to create a process



Steps to create a process

The screenshot displays the UiPath Studio Community interface. The main workspace shows a 'Main Sequence' with a 'Raise Alert' activity. The 'Raise Alert' activity configuration is as follows:

- Orchestrator Folder Path: thebest.divya@gmail.com's workspace
- Notification: Your memory is full
- Severity: Info

A 'Configure it' dialog box is overlaid on the workspace, displaying the message: 'Make sure you configure the required activity fields and there are no errors in your activity before moving forward.' The dialog box has 'Previous' and 'Next' buttons. The 'Next' button is highlighted.

The interface also shows the 'Activities' pane on the left, the 'Data Manager' pane on the right, and the 'Output' pane at the bottom. The status bar at the bottom indicates the user is logged in as 'thebest.divya@gmail.com's...' and the system time is 09:22 on 06-02-2024.

Steps to create a process

The screenshot displays the UiPath Studio Community interface. The main workspace shows a 'Main Sequence' with a 'Raise Alert' activity. The 'Raise Alert' activity configuration window is open, showing the following fields:

- Orchestrator Folder Path: thebest.divya@gmail.com's workspace
- Notification *: Your memory is full
- Severity *: Info

A 'Configure it' dialog box is overlaid on the right, with the text: 'Make sure you configure the required activity fields and there are no errors in your activity before moving forward.' The dialog shows 'Step 2 of 4' and has 'Previous' and 'Next' buttons.

The interface includes a top menu bar with 'File', 'HOME', 'DESIGN', and 'DEBUG'. The 'DESIGN' tab is active. The left sidebar shows the 'Activities' pane with a search bar and a list of activities, including 'Raise Alert'. The right sidebar shows the 'Data Manager' pane with a search bar and a list of data types, including 'Variables', 'Arguments', 'Constants', 'Resources', and 'Entities'. The bottom status bar shows the user 'thebest.divya@gmail.com's...' and the window title 'VB, Windows'.

Steps to create a process

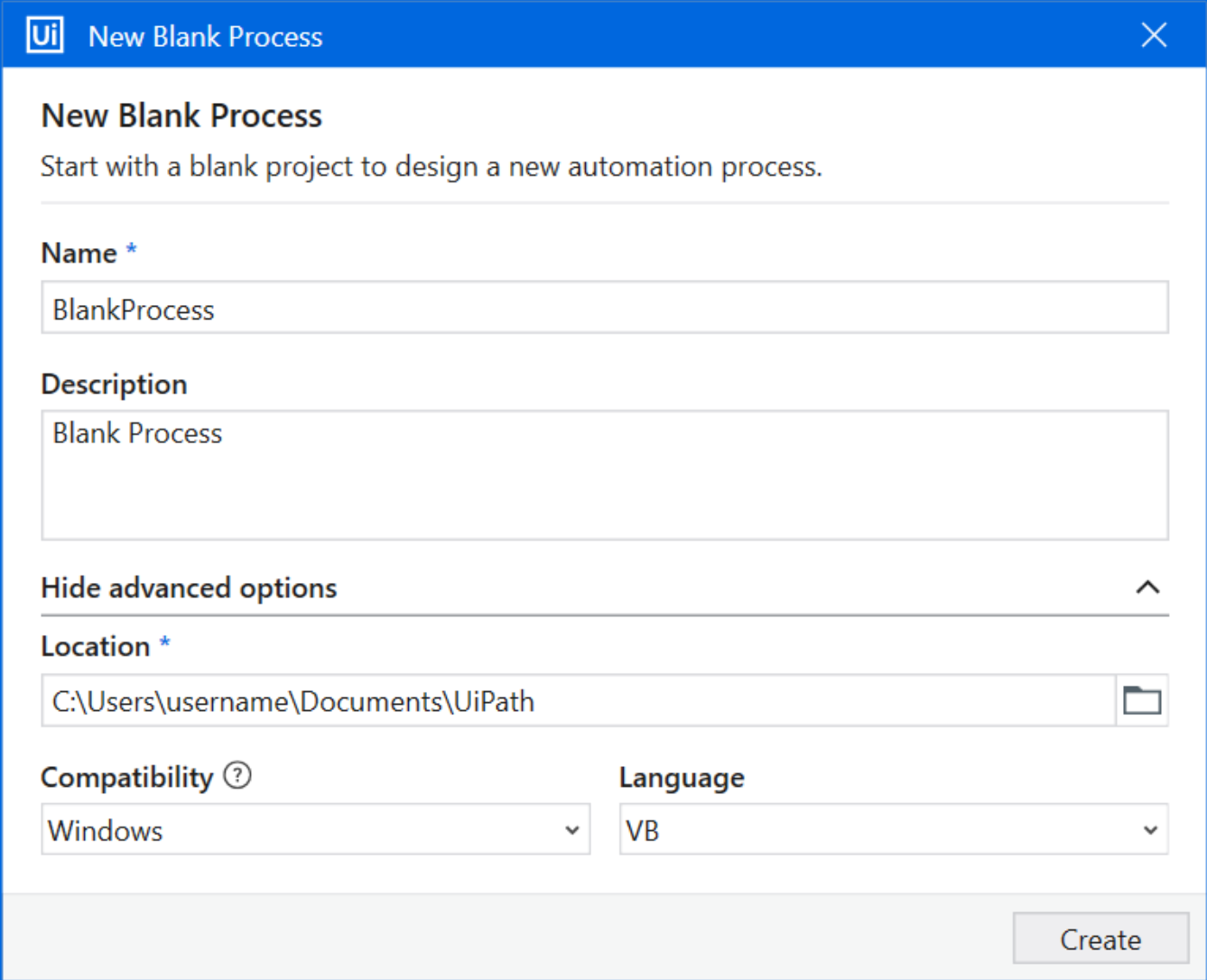
The screenshot displays the UiPath Studio Community interface. The top menu bar includes 'File', 'HOME', 'DESIGN', and 'DEBUG'. The 'DESIGN' tab is active, showing a 'Main' sequence. A 'Raise Alert' dialog box is open, prompting for an 'Orchestrator Folder Path' (set to 'thebest.divya@gmail.com's workspace'), a 'Notification' (set to 'Your memory is full'), and a 'Severity' (set to 'Info'). Below this, a message box states: 'Now continue adding activities. Add activities faster by clicking on the drop area. This will trigger the search menu.' The 'Data Manager' pane on the right lists various data types like Variables, Arguments, Constants, Resources, Entities, Processes, Assets, Queues, Connections, and Namespaces. The bottom status bar shows 'Step 3 of 4' and a 'Next' button. The Windows taskbar at the bottom includes the search bar, task icons, and system tray information (10°C Fog, 09:22, 06-02-2024).

Steps to create a process

The screenshot displays the UiPath Studio Community application. The top menu bar includes 'File', 'HOME', 'DESIGN', and 'DEBUG'. The 'DEBUG' tab is active, showing a toolbar with icons for 'New', 'Save', 'Export As...', 'Debug File', 'Cut', 'Copy', 'Paste', 'Undo', and 'Redo'. Below the toolbar, the 'Main' workspace shows a 'Main Sequence' diagram. A 'Raise Alert' dialog box is open, prompting for 'Orchestrator Folder Path' (set to 'thebest.divya@gmail.com's workspace'), 'Notification *' (set to 'Your memory is full'), and 'Severity *' (set to 'Info'). The left sidebar contains a 'Recent' list with categories like 'Installed', 'Alerts', 'API', 'App Integration', 'Assets', 'Collection', 'Computer Vision', 'Data', 'Formatting', 'Integrations', 'Jobs', 'Logging', and 'Orchestrator'. The right sidebar shows a 'Data Manager' pane with a search bar and a list of data types including 'Variables', 'Arguments', 'Constants', 'Resources', 'Entities', 'Processes', 'Assets', 'Queues', 'Connections', and 'Namespaces'. The bottom status bar indicates the user is 'thebest.divya@gmail.com's...' and the application is running on 'VB, Windows'. The Windows taskbar at the bottom shows the search bar, task view, and various application icons, along with system information like '10°C Fog' and '09:22 06-02-2024'.

Creating a Basic Process

Step 1: Launch Studio. In the HOME Backstage view, click **Process** to create a new project. The **New Blank Process** window is displayed



Ui New Blank Process

New Blank Process
Start with a blank project to design a new automation process.

Name *
BlankProcess

Description
Blank Process

Hide advanced options ^

Location *
C:\Users\username\Documents\UiPath

Compatibility ?
Windows

Language
VB

Create

Creating a Basic Process

Step 2:

In the New Blank Process window:

- Enter a name for the new project and a description that summaries what you are aiming to do with this automation project.
- Select **Show advanced options** and choose the location where to create the project.
- Select **Windows** for the **Compatibility** option and **VB** for **Language**.
- Click on Create option

UiPath New Blank Process

New Blank Process
Start with a blank project to design a new automation process.

Name *
BlankProcess

Description
Blank Process

Hide advanced options ^

Location *
C:\Users\username\Documents\UiPath

Compatibility ? Windows ▾ **Language** VB ▾

Create

Creating a Basic Process

Step 3: Open Internet Explorer and navigate to <https://www.uipath.com/product/studio>. You can use any supported browser, but make sure the required [UiPath extension](#) is installed.

Step 4: In the **Activities** panel, search for [Use Application/Browser](#) and drag it to the **Designer** panel. Notice that the activity is automatically added inside a **Sequence**.

Step 5: In Use Application/Browser, click **Indicate application to automate**, and then move the mouse pointer to the browser page you just opened. When the window is highlighted, click anywhere in the browser page.

The Use Application/Browser activity is updated, the URL is added to the **Browser URL** field, and a screenshot of the window appears inside the activity.

Creating a Basic Process

Step 6: From the **Activities** panel, add a [Get Text](#) activity to the **Do** container of the **Use Application/Browser** activity. Click **Indicate in IE**, and click on the text that you want to scrape from the previously loaded web page.

Step 7: In the [Variables](#) panel, create a new string variable called GetText, and set its scope to Sequence. The purpose of this variable is to store the scraped information.

Creating a Basic Process

Step 8: In the **Get Text** activity, insert the GetText variable in the **Save to** field.

Step 9: From the **Activities** panel, drag a [Write Line](#) activity. In the input text field, add the previously created variable. This activity writes the scraped data to the **Output** panel.

Name	Variable type	Scope	Default
GetText	String	Sequence	<i>Enter a VB expression</i>
<i>Create Variable</i>			
Variables	Arguments	Imports	  100%   

Creating a Basic Process

Sequence

This simple example opens Internet Explorer, goes to a certain web page, scrapes a certain text container, writes the text to the Output panel, and to a Notepad file.

Use Browser IE: Advanced Automation Software, RPA Tools - Studio | Ui

This activity opens Internet Explorer and navigates to the specified web page.



Browser URL

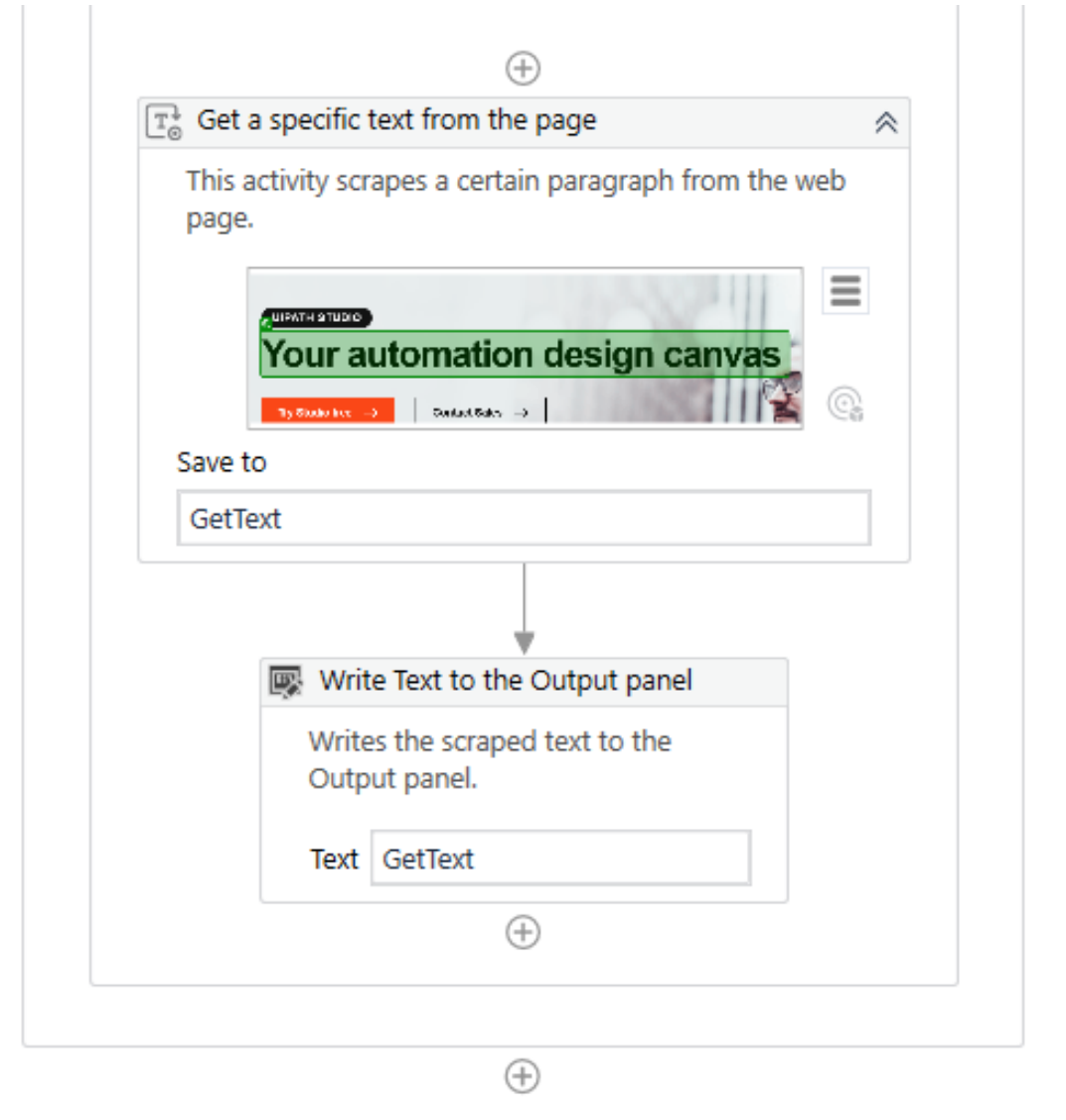
"https://www.uipath.com/product/studio"

Do

This container has activities for scraping text, writing to console, and to a Notepad file.

Get a specific text from the page

Dr. Divya



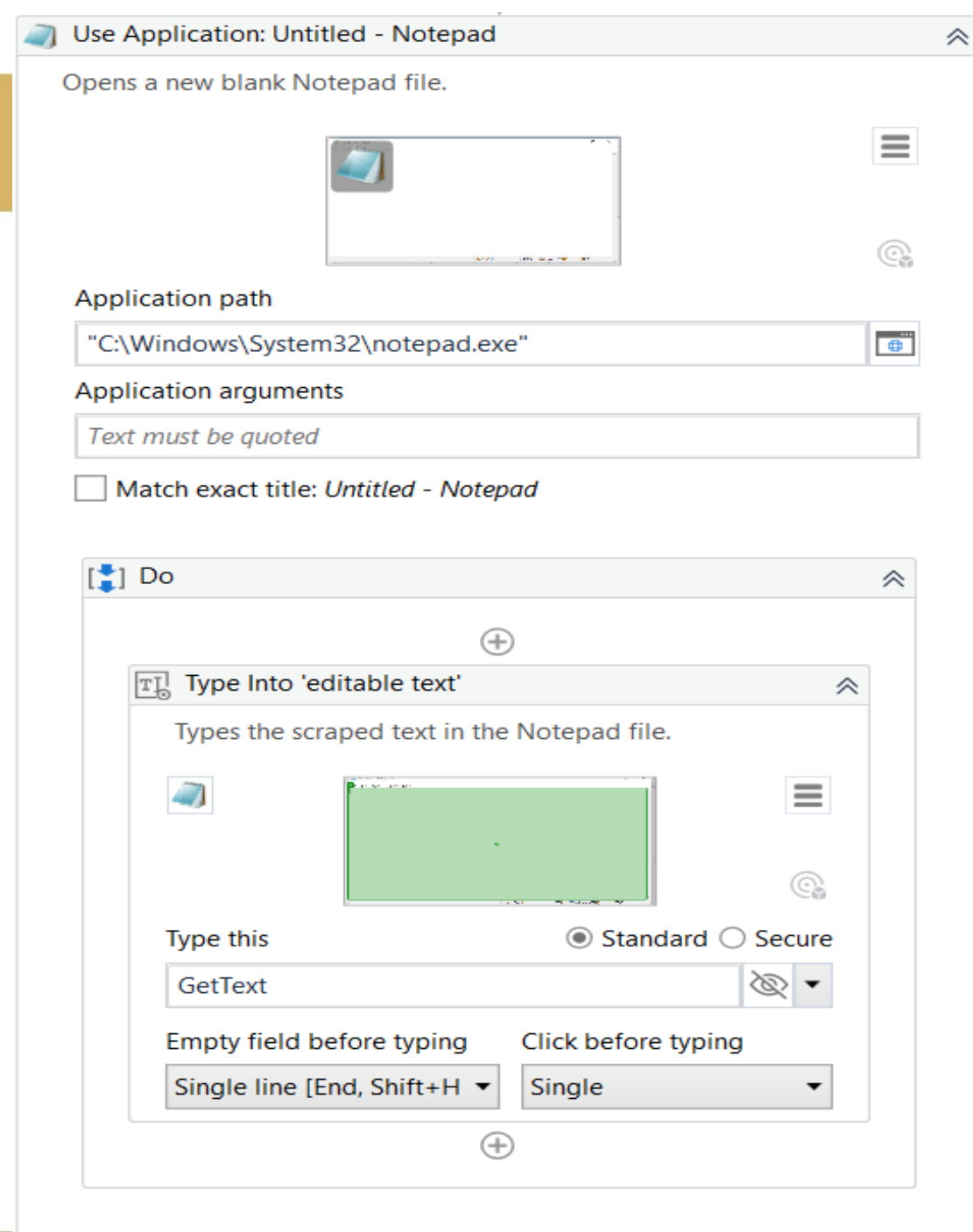
Creating a Basic Process

Step 10: Add another **Use Application/Browser** activity to the project. Open a Notepad instance, click **Indicate application to automate**, and then move the mouse pointer to the Notepad window you just opened. When the window is highlighted, click anywhere inside it.

The Use Application/Browser activity is updated, the path is added to the **Application path** field, and a screenshot of the window appears inside the activity.

Creating a Basic Process

Step 11: Add a [Type Into](#) activity in the **Use Application/Browser** activity's **Do** container. Click **Indicate in App** to select the Notepad window, and add the GetText variable to the **Type this** field. This activity writes the scraped text to a Notepad window.



Creating a Basic Process

Step 12: In the ribbon, click **Run File** or use the **Ctrl + F6** shortcut to execute the project.

- When the execution is completed, notice that the scraped text is written to the [Output](#) panel, and in the Notepad file.
- Similarly, you can create other processes to scrape data from web pages, using the [Table Extraction](#) wizard.

- **UNIT I**

□ Introduction to process automation: Definition of automation, Socio economic impacts of automation, Types of Automation, Introduction to Robot Process Automation, How is Automation Driving the Digital Transformation? Role of Automation in Business; Build first automation process with UiPath StudioX